

The Eyes of Texas are Upon OB/GYNs: Physician Migration and Crowdsourced Enforcement of Abortion Regulations ^{*}

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Abstract

This paper studies the labor market effects of civil liability enforcement using Texas Senate Bill 8 (SB-8, 2021) as a natural experiment. A civil liability enforcement mechanism in this context allows any citizen to sue anyone who “aids or abets” a prohibited abortion for minimum damages of \$10,000. This “bounty law” lowers the evidentiary burden and expands the pool of potentially liable individuals, which raises expected operating costs and could incentivize out-migration for a broad set of physicians compared to traditional criminal enforcement. We exploit variation across physician specialties, states, and time in a triple-difference design, comparing reproductive health physicians to all other physicians in Texas against similarly abortion-hostile states before and after SB-8’s signing. Using a Medicare administrative provider panel of approximately 1.4 million physicians (2007–2022) and Wagescape job posting data (2016–2022), we find that reproductive health physicians did not leave Texas at differential rates following SB-8. The null result is consistent with the high fixed cost of interstate relocation, estimated at \$150,000 to \$250,000. Neither posted salaries nor job posting volume for reproductive health providers changed differentially. Our findings are corroborated by synthetic difference-in-differences and synthetic control estimators and provide a contrast to prominent media narratives of a large OB/GYN exodus from Texas. We contribute to a rapidly emerging literature by providing evidence that civil liability enforcement had no detectable effect on the Texas physician labor market above and beyond the broader abortion policy environment.

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1 Introduction

Texas Senate Bill 8, enacted in September 2021, introduced a novel civil liability enforcement mechanism (CLEM). Unlike all other abortion restrictions enforced through criminal penalties, SB-8 authorized “any person” to bring civil lawsuits against anyone who “aids or abets” an abortion after detection of a fetal heartbeat (around six weeks of gestation). The so-called “bounty law” could have been challenged in the courts, but the civil liability enforcement mechanism was included for the express purpose of helping Senate Bill 8 survive judicial review.

Civil liability enforcement potentially imposes different deterrent effects and risk compared to criminal prosecution. Criminal liability requires a burden of proof that is “beyond a reasonable doubt,” but civil liability only requires a “preponderance of the evidence.”¹ Imagine the less politically charged example of shoplifting. If the state uses traditional criminal liability, a store in that state may accuse a customer of shoplifting. Police would gather evidence and may decide to make an arrest. The customer would be arraigned before a grand jury who may decide there is enough evidence to continue with prosecution. Finally, the customer would stand before a judge the determination of their guilt. If the state used civil liability to enforce its shoplifting law, anyone could accuse anyone of “aiding or abetting” shoplifting by filing a lawsuit for a minimum of \$10,000 in damages. The accused would be responsible for their own legal fees and cannot recover their legal fees if exonerated. Criminal liability imposes a cost on those intending to shoplift, but civil liability imposes a cost on every customer in the store. It would be reasonable for a shopper to avoid shopping in a state with civil liability enforcement.

[Becker \(1968\)](#) establishes the economic framework: physicians weigh the expected benefits against the expected costs of practicing in Texas. Expected costs in this context include the penalty for violating Texas’s abortion law multiplied by the probability of detection

¹A preponderance of evidence means there is more than a fifty percent chance that the party being sued is at fault. Proof beyond a reasonable doubt means that a reasonable person would have no doubt about the defendant’s guilt.

and conviction. Under criminal enforcement of abortion restrictions, that expected cost would fall exclusively on the physician who performs the procedure. Physicians who do not directly perform abortions would not weight that expected cost as part of their location decision. Under SB-8’s civil enforcement, the expected cost changes in two important ways. First, the “preponderance of the evidence” standard raises the probability of liability relative to criminal prosecution. Second, the “aid or abet” language extends potential liability from exclusively abortion providers to any physician who manages pregnancy complications, counsels patients about their options, or provides emergency obstetric care. The result is an increase in expected cost that applies even to physicians who never perform an abortion. This framework predicts that the structure of SB-8 increased expected operating costs for reproductive health physicians (especially OB/GYNs, family medicine physicians, and emergency medicine physicians) and generated direct financial incentives to exit Texas.

This paper examines migration responses to civil liability enforcement mechanisms in the context of reproductive health physicians responding to Texas Senate Bill 8. SB-8’s signing in May 2021 and implementation in September 2021 provide a natural experiment with one treated state. SB-8 took effect nine months before *Dobbs v. Jackson Women’s Health Organization* overturned the constitutional right to an abortion from *Roe v. Wade*. This context differs markedly from the post-*Dobbs* period when multiple states enacted abortion regulations. An affected physician had greater ability to move to unaffected locations during the SB-8 era than in the post-*Dobbs* period, when abortion restrictions spread across many states simultaneously.

We use two complementary data sources. We first use Medicare administrative data on physician locations from November 2007 through June 2022 (before the *Dobbs* ruling), covering approximately 1.4 million physicians, to estimate triple-difference models comparing reproductive health physicians (OB/GYNs and family medicine physicians) to all other physicians in Texas relative to other states, before and after SB-8’s signing on May 19,

2021.² Physician counts are normalized per 100,000 women of childbearing age. We also use Wagescape data on physician job postings covering 1,327,394 postings from 2016–2022, including 107,474 OB/GYN and 150,839 family medicine postings. The posted salary is available for approximately half of postings. These data allow us to examine whether employer demand and compensation for reproductive health physicians in Texas shifted following SB-8.

We find no statistically significant migration response for reproductive health physicians in the thirteen months following SB-8’s signing. The Wagescape job posting data reinforce this null: both posted salaries and job posting volume for Texas reproductive health providers show no differential change relative to controls. When the sample is expanded to include all family medicine physicians, a salary decline appears in hospitalist and “without OB” postings, the subset least plausibly exposed to SB-8’s “aid or abet” language, suggesting employer composition effects rather than a CLEM wage response. The null results across both data sources are consistent with the high fixed costs of interstate practice relocation. These findings are inconsistent with prominent media narratives characterizing the post-SB-8 period as an exodus of OB/GYNs from Texas ([Petterson and Eubank, 2024](#); [Taladrid, 2024](#)); our results do not support the conclusion that a large-scale departure occurred.

A substantial literature documents that criminal abortion restrictions reduce the supply of reproductive health physicians, with effect sizes scaling in the severity of enforcement. In the TRAP era predating *Dobbs*, [Chatterji et al. \(2025\)](#) find a 6.6 percent reduction in OB/GYN supply following targeted abortion provider regulations using a stacked difference-in-differences design, with evidence pointing to workforce exit rather than geographic relocation. [Markowski and Vandenbroeck \(2025\)](#) estimate a 4.67 percent reduction in OB/GYN density over 1993–2021 under TRAP laws, with effects persisting for a decade in nonmetropolitan counties. The effects of criminal restrictions extend beyond the current workforce to the training pipeline: [Hammoud et al. \(2024\)](#) document a significant decline in

²We measure treatment from SB-8’s signing rather than its September 1, 2021 enactment date because Google Trends data on searches for “Texas Heartbeat” show a large spike in attention at the time of signing. This response suggests that Senate Bill 8 was highly salient at its signing.

residency applications to OB/GYN programs in abortion-restrictive states following *Dobbs*.

In the post-*Dobbs* era, criminal abortion bans have produced larger and faster responses. [Rader et al. \(2025\)](#) document a 35 percent loss of practicing OB/GYNs in Idaho within 27 months of a near-total criminal ban with felony charges – the sharpest estimate in the literature. [Zhu et al. \(2025\)](#) find a 4.2 percent decrease in OB/GYN practitioners per 100,000 reproductive-aged women in the most restrictive states, and [Diaz-Campo and Pineda-Torres \(2026\)](#) find a 3 percent decline across ban states. Across settings, effect sizes appear to scale with the severity and certainty of criminal liability, suggesting it is an important driver of the exit margin.

Against this backdrop, several studies find null or modest migration effects for established physicians in the short run. [Staiger et al. \(2025\)](#) study practice locations for 60,085 OBGYNs across all policy environments from January 2018 to September 2024 and find no statistically significant change in counts across banned, threatened, and protected states, with approximately 95 percent remaining in their pre-*Dobbs* policy environment. [Strasser et al. \(2024\)](#) similarly find no significant post-*Dobbs* shift in new-state enrollment flows comparing ban and no-ban states. [Nelson and Witko \(2026\)](#) document a preference-behavior gap: physicians in survey experiments express a preference for non-ban states, but actual migration effects remain modest. This behavioral stickiness is consistent with high fixed costs of relocation: [Falcettoni \(2021\)](#) identifies strong location preferences near residency training as a primary driver of physician location choice, and [Dench et al. \(2025\)](#) find significant out-migration only from states with total criminal bans, not from states with partial restrictions. No prior work has examined whether civil liability enforcement, which broadens the set of physicians facing expected operating costs, produces a different migration equilibrium.

2 Background: SB-8 and Civil Liability Enforcement

Texas Senate Bill 8 (SB-8) was signed on May 19, 2021 and enacted on September 1, 2021. SB-8 prohibits abortion after the detection of embryonic cardiac activity which typically occurs around six weeks of gestation. The statute’s enactment predated the Supreme Court’s decision in *Dobbs v. Jackson Women’s Health Organization* by nine months, placing it within the constitutional framework established by *Roe v. Wade* and *Planned Parenthood v. Casey*. Under precedent, SB-8 could have been ruled unconstitutional. The statute survived pre-enforcement judicial review through an unprecedented enforcement mechanism that delegated enforcement exclusively to private civil litigation rather than state officials. Chief Justice John Roberts characterized this approach as “not only unusual, but unprecedented,” warning that “it is the role of the Supreme Court in our constitutional system that is at stake.” SB-8 avoided constitutional challenge by eliminating any state official against whom plaintiffs could seek injunctive relief.

The timing and scope of SB-8 merit particular attention for understanding its impact on physician practice patterns. At the time of enactment, approximately 85-90% of abortions performed in Texas occurred after six weeks of gestation, meaning the statute effectively prohibited the vast majority of abortion procedures previously performed in the state. Unlike subsequent abortion restrictions that took effect following *Dobbs*, SB-8 operated in a legal environment where abortion remained constitutionally protected nationally, creating a localized regulatory shock specific to Texas providers and patients.

SB-8 authorizes “any person, other than an officer or employee of a state or local governmental entity,” to bring civil action against anyone who performs or “aids or abets” an abortion prohibited under the statute. This “aid or abet” language extends potential liability beyond physicians performing procedures to encompass a broad range of supportive activities, including providing transportation, financial assistance, counseling, or administrative support for prohibited abortions.

The incentives embedded in CLEM substantially favor plaintiffs. Successful litigants

receive minimum statutory damages of \$10,000 per abortion plus injunctive relief and attorney’s fees. Defendants cannot recover attorney’s fees or costs even if they win in litigation, creating asymmetric exposure to legal expenses. This one-way cost structure distinguishes CLEM from typical civil litigation, where prevailing defendants may recover costs, and from criminal prosecution, where defendants face no financial penalty for successful defense beyond legal fees. The statute further advantages plaintiffs by eliminating the heightened burden of proof required in criminal proceedings; civil litigation requires only a preponderance of evidence rather than proof beyond reasonable doubt.

A single prohibited abortion could generate multiple lawsuits from different plaintiffs, each seeking the \$10,000 statutory minimum. For potential defendants, the expected cost of CLEM exposure reflects not only the probability of liability but also the probability of facing litigation itself, multiplied by defense costs that cannot be recovered. Medical malpractice insurance does not cover civil liability. SB-8 litigation imposes defense costs with no possibility of fee recovery and no insurance coverage.

Despite the statute’s theoretical enforcement potential, actual litigation activity under SB-8 was limited. Only three lawsuits were filed pursuant to the statute, all between September 20–22, 2021, and all targeting a single physician: Dr. Alan Braid of San Antonio. Dr. Braid had deliberately invited litigation by publishing an op-ed in *The Washington Post* announcing that he had performed an abortion in violation of SB-8. No additional lawsuits have been filed under SB-8’s provisions, no plaintiff has collected the statutory bounty, and no case has resulted in a finding of liability. The sole case to reach judicial resolution, *Gomez v. Braid*, was dismissed for lack of standing in December 2022 and affirmed on appeal in February 2024, establishing that the statute’s “any person” language remains subject to Texas constitutional standing requirements.

The absence of broader enforcement activity reflects strategic inaction by abortion opponents rather than lack of interest or resources. Major anti-abortion organizations, including Texas Right to Life and Operation Rescue, explicitly declined to file SB-8 lawsuits to avoid

the creation of test cases that could provide grounds for invalidating the statute. By maintaining the threat of enforcement while limiting actual litigation, these organizations preserved SB-8’s deterrent effect while minimizing legal risk to the statute. Abortion providers in Texas ceased performing procedures after detection of embryonic cardiac activity upon SB-8’s effective date, consistent with deterrence operating through the threat of litigation rather than its enforcement.

The persistent threat of litigation inherent in CLEM, whereby any patient, family member, or stranger may initiate suit, differs from criminal enforcement risk in a key way. Criminal prosecution requires a willing prosecutor; CLEM requires only a willing plaintiff for whom the statute creates affirmative financial incentives and eliminates downside risk. For physicians evaluating practice location, CLEM exposure represents an uninsurable increase in expected operating costs.

3 Conceptual Framework

Becker’s (1968) model of rational deterrence treats the decision to engage in a regulated activity as a function of expected liability costs: the probability of detection multiplied by the penalty conditional on detection, weighed against the expected benefit of the activity. SB-8 is designed to alter both components of this expected cost in ways that are qualitatively distinct from traditional criminal enforcement.

Probability of detection. Criminal enforcement targets the physician who performs the procedure, requiring that a state actor identify, investigate, and prosecute a specific individual. SB-8’s “aid or abet” language dramatically broadens the set of actors subject to potential liability. A physician who provides standard obstetric care or counseling to a patient who subsequently obtains an abortion may be named in a civil suit without having performed or directly facilitated an abortion. The vagueness of the statute’s scope means that even a physician taking no deliberate part in an abortion faces a positive and difficult-

to-quantify probability of being drawn into litigation. This ambiguity itself constitutes a cost: physicians cannot insure against risks they cannot precisely estimate.

Penalty structure. Under criminal enforcement, a defendant who prevails at trial bears no financial penalty beyond legal fees typically covered by malpractice insurance. SB-8 inverts this logic. Defendants cannot recover attorney’s fees or costs even if they prevail in litigation, meaning a successful defense still imposes a net financial cost. Moreover, a single prohibited procedure can generate multiple independent lawsuits from different plaintiffs, each seeking the \$10,000 statutory minimum, creating effectively unbounded aggregate exposure from a single clinical event. Because standard medical malpractice policies do not cover SB-8 civil liability, these costs are uninsurable.

Together, the broader detection margin and the asymmetric penalty structure imply that SB-8 imposes expected costs on a much wider set of physicians than traditional criminal enforcement, including physicians who provide no abortion services but whose routine clinical activities might be characterized as “aiding” a prohibited abortion. The predicted behavioral response is either to relocate away from Texas (the migration margin) or to demand higher compensation for bearing this incremental risk (the wage margin). The evidence in this paper finds no detectable response on either margin, consistent with the high fixed costs of interstate practice relocation estimated at \$150,000–\$250,000.

4 Data

4.1 Sample Group: “Hostile” and “Restricting” States

Texas was not the only state facing a restrictive abortion environment during the analysis period. Using all states as the control group would contaminate the comparison: physicians in states with open abortion laws are operating in a fundamentally different policy atmosphere compared to a physician in a state that restricts abortion. To address this, we restrict the sample group to the twenty-five states that [Dench et al. \(2025\)](#) classify as “hostile to

abortion” or subject to a total abortion ban. The policy environment in these states most closely resembles Texas’s policy environment before and around SB-8.³

This restriction serves a substantive identification purpose beyond improving pre-period balance. Physicians practicing in hostile or ban states already operate under a restrictive abortion environment. A triple-difference comparison of Texas RHPs against RHPs in these states therefore isolates the *marginal* effect of SB-8’s civil liability enforcement mechanism above and beyond the effect of a restrictive abortion policy environment. By contrast, a comparison against physicians in permissive states would conflate the CLEM-specific deterrent with the broader effects of practicing in any abortion-restrictive state.

Table 2 confirms that the restricted sample is more similar to Texas on pre-period physician market outcomes than the full pool of states.

4.2 Migration Data

This paper uses Medicare administrative data to construct a quarterly panel tracking physician practice locations over time. The primary data sources are the National Plan and Provider Enumeration System (NPPES) and the Centers for Medicare and Medicaid Services (CMS) Doctor and Clinician (DAC) database. The NPPES serves as the authoritative registry of all healthcare providers in the United States who hold a National Provider Identifier (NPI), a unique identification number required for billing Medicare and private insurance. The DAC database supplements NPPES by providing detailed practice location information for physicians who bill Medicare. Because practice addresses must be current to receive Medicare reimbursement, these administrative records offer high-quality location data with strong incentives for accuracy and timeliness.

The sample spans November 2007 through June 2022, creating a quarterly panel that covers approximately 1.4 million physicians. This period begins with the earliest available DAC data and concludes with the Supreme Court’s *Dobbs* decision in June 2022. The panel

³The 25 control states are: AL, AR, AZ, FL, GA, IA, ID, IN, KY, LA, MO, MS, NC, ND, NE, OH, OK, PA, SC, SD, TN, UT, WI, WV, and WY.

includes both physicians already in practice and new physicians entering the workforce from residency programs, providing broad coverage of the physician labor market over the study period. For each physician, the data contain the provider’s name, NPI, practice street address, and specialty designation.

Five dependent variables capture different dimensions of physician mobility. First, a physician is coded as “active in the state” if she appears in the data with a practice location in that state during a given quarter. Second, a physician is classified as having “exited the state” if she leaves for another state while remaining in active practice. Third, a physician “entered the state” from another state while remaining in active practice. Fourth, “new entrance” identifies each physician’s very first active quarter in the data, capturing a new residency graduate making their entry into the workforce. Fifth, “retirement” captures permanent career exit by identifying each physician’s final active quarter. Summary statistics are shown in Table 1.

	All	TX RHP	TX Non-RHP	Non-TX RHP	Non-TX Non-RHP	(Min, Max)
<i>Panel A: NPI Physician Panel (per 100k women)</i>						
Active Physicians	826 (396)	365 (42.50)	1017 (117)	481 (89.07)	1182 (251)	(256, 2050)
Exit State	6.29 (11.22)	1.39 (2.26)	5.88 (9.12)	3.05 (4.63)	9.76 (14.61)	(0, 140)
Entry State	6.57 (11.88)	1.69 (2.08)	6.77 (8.25)	3.02 (4.82)	10.30 (15.50)	(0, 176)
New Career Entrants	15.04 (80.62)	7.08 (31.68)	19.08 (87.27)	9.14 (43.75)	21.08 (106)	(0, 1226)
Retirements	2.30 (15.50)	1.19 (7.61)	2.77 (17.35)	1.57 (10.31)	3.05 (19.45)	(0, 296)
<i>Panel B: Wagescape Job Postings</i>						
Posted Salary (USD, COL-Adj.)	184,510 (116,651)	150,084 (104,744)	180,711 (106,084)	175,857 (115,855)	187,716 (118,228)	(10,266, 863,998)
Log Posted Salary	11.93 (0.689)	11.64 (0.818)	11.94 (0.633)	11.83 (0.772)	11.95 (0.671)	(9, 13.67)
Job Postings (per 100k women 15–44)	46.13 (68.27)	8.82 (6.89)	37.65 (28.70)	19.37 (26.34)	74.69 (85.89)	(0.1, 927)

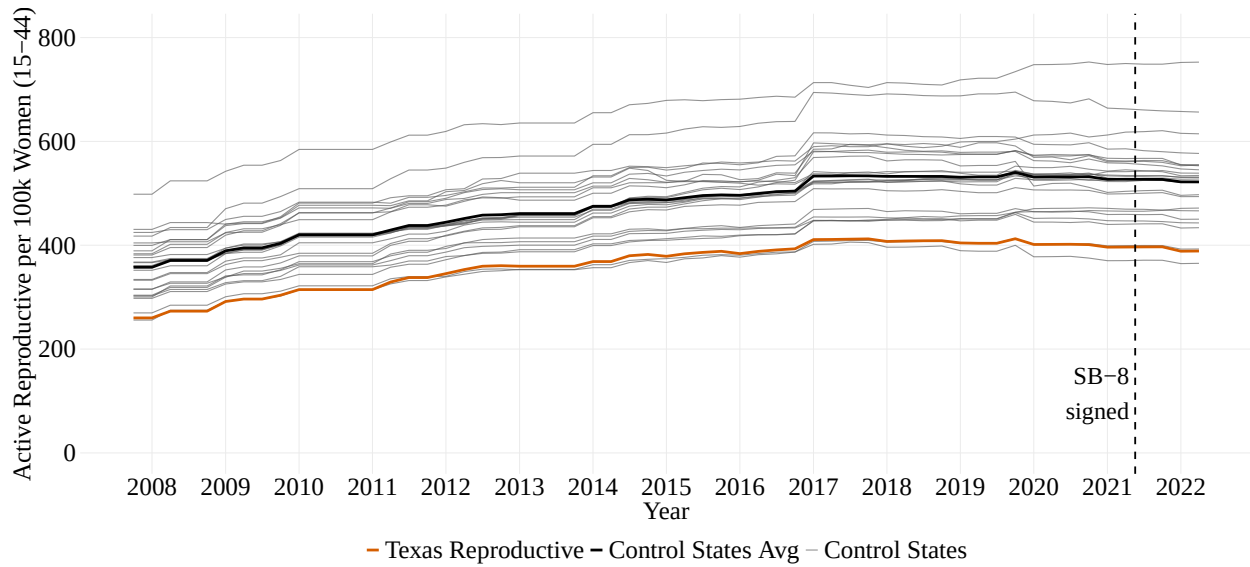
Table 1: Summary statistics for both data sources. Columns show means by group: all observations, Texas RHP, Texas non-RHP, non-Texas RHP, non-Texas non-RHP (restricted control states), and overall (Min, Max). SD in parentheses below means. Panel A: NPI physician panel; unit is the state $\times$ quarter cell collapsed to RHP and non-RHP groups; outcomes per 100,000 women of childbearing age. Panel B: Wagescape job postings restricted to the analysis sample; salary unit is the individual posting; postings unit is the state $\times$ quarter cell. Sources: NPPES/CMS DAC; Wagescape via Dewey Data Platform.

Physician specialties are grouped into *reproductive health providers* (RHPs) and all other physicians. RHPs include obstetricians and gynecologists (OB/GYNs), family medicine physicians, and emergency medicine physicians. OB/GYNs face the most direct exposure

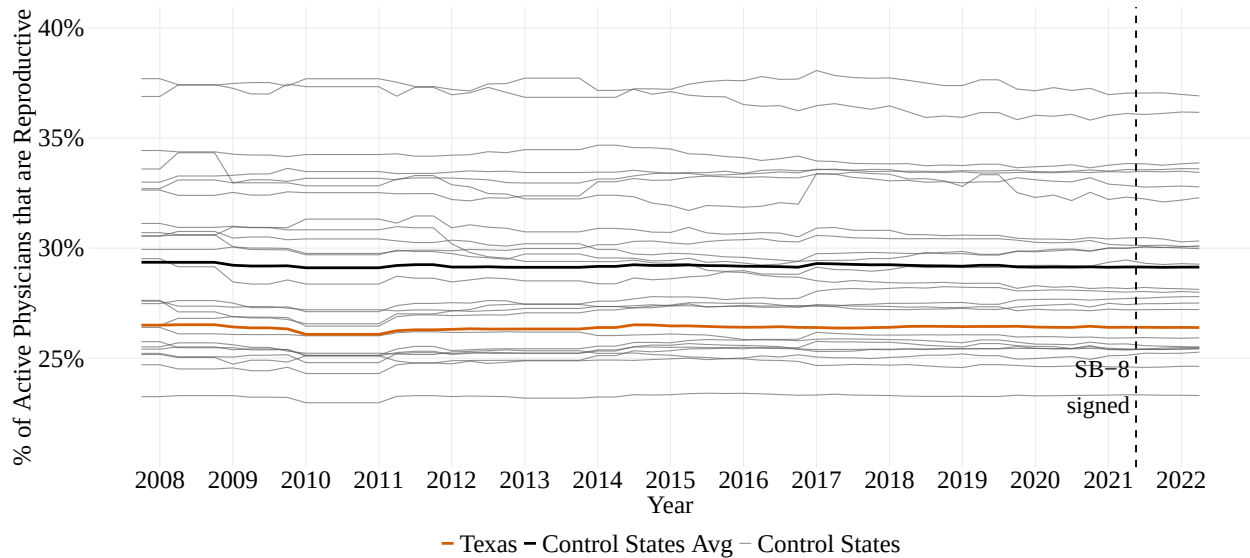
to SB-8’s provisions. Family medicine physicians are included because they may serve as the sole maternal healthcare provider in rural communities and encounter pregnancy complications as part of primary care. Emergency medicine physicians are included because they face time-sensitive decisions in obstetric emergencies where the line between permissible and impermissible treatment is ambiguous under the statute’s “aid or abet” language. All outcomes are normalized per 100,000 women of childbearing age in each state-quarter to account for the substantial variation in population across states.⁴

Figure 1 shows active reproductive health providers (RHPs) per 100,000 women of childbearing age and RHPs as a share of all active physicians. Each figure shows Texas (bold orange) alongside the average of the restricted control states (bold black) and the trajectory of each individual control state (thin gray lines). The Texas series tracks the control average closely throughout the pre-period and shows no visible divergence following SB-8’s signing in May 2021, consistent with the null result.

⁴Women of childbearing age are defined as ages 15–44, following standard practice in the obstetrics literature.



(a) Active RHPs per 100k women: Texas vs. control states.



(b) RHPs as a share of all active physicians (%): Texas vs. control states.

Figure 1: Active reproductive health provider (RHP: OB/GYN, Family Medicine, and Emergency Medicine) counts per 100,000 women of childbearing age (ages 15–44) from 2007 to 2022. Panel (a) plots RHP counts per 100k women; panel (b) plots the RHP share of all active physicians. Both panels show Texas (bold orange), the average of the hostile/ban control states (bold black), and each individual control state (thin gray lines). The dashed vertical line marks May 19, 2021, the date SB-8 was signed. Source: NPPES/CMS DAC.

The universal coverage and longitudinal structure of these data make them well-suited for analyzing physician location decisions in response to state-level policy changes, and the administrative nature of the data reduces concerns about survey non-response or selective

attrition. The resulting data are a panel of specialty-state-quarter counts of migration, normalized per 100,000 women of childbearing age.

A limitation in our data is that we observe a lower bound on physician retirements. In general, most physicians do not deactivate their NPIs when they retire, so we substantially undercount true retirements. The retirement margin is particularly important for our analysis because physicians planning to retire in the near future may exit early rather than bear the legal risk of continued practice under SB-8, and the fixed cost of retiring a few years early is lower than the fixed cost of relocating for mid-career physicians. We are seeking methods to bolster our measurement of physician retirement, and a further discussion of this point is included in the limitations and next steps at the end of this paper.

4.3 Wagescape Job Postings Data

We use physician job posting data from Wagescape to analyze the offered salary and job posting volume for reproductive health physicians. These data cover 1,327,394 physician job postings for the period 2016–2022, including 107,474 OB/GYN postings, 150,839 Family Medicine postings, and 49,068 Emergency Medicine postings.⁵ Each posting records the position state, date posted, and, if disclosed by the employer, the offered salary. Employers disclose salary in approximately 50 percent of postings, with coverage rising from 5–13 percent in 2016–2017 to 65–80 percent in 2018–2022. After limiting to the 50 US states and the District of Columbia, the salary analysis sample includes all postings with a positive reported salary across all three physician specialties in the RHP group.

A potential threat to identification in the wage analysis is differential trends in salary

⁵OB/GYN postings are identified using SOC code 29-1218. Family medicine postings are identified using SOC code 29-1215, and we remove Family medicine postings that include “no OB” in the job title. ER physicians are identified using the job title keywords “emergency medicine,” “emergency physician,” “ER physician,” “EM physician,” and “ED physician.” The NPI analysis classifies all Family Medicine physicians (taxonomy 207Q) as reproductive health providers, while the Wagescape wage analysis restricts to Family Medicine physicians whose job title indicates obstetric, emergency, or otherwise ambiguous reproductive content. In the NPI data, it is not possible to identify which family physicians perform obstetric services from the administrative records alone. This means the NPI RHP group is broader than the Wagescape RHP group.

disclosure rates across groups. If Texas RHP employers became more or less likely to disclose posted salaries around SB-8’s signing (relative to Texas non-RHPs or to control-state RHPs) the estimated wage effect could reflect a composition change in the salary-reporting sample rather than a true price signal. Panel (a) of Figure 2 plots the quarterly share of postings with a salary listed for four groups: Texas RHP, control-state RHP, Texas non-RHP, and control-state non-RHP. The four series track closely throughout the sample period, including around the May 2021 signing date, with no visible differential break for Texas RHPs relative to any comparison group. This parallel behavior in disclosure rates supports the interpretation that changes in the salary sample composition are not a confounding factor in the wage triple-difference estimates.

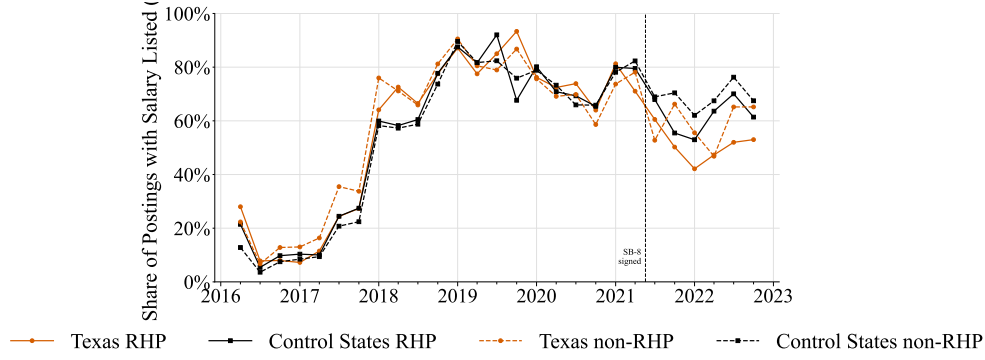
Panel (b) shows the total volume of physician job postings per 100,000 women of child-bearing age, comparing Texas to the control states for all physicians and for RHPs separately. Across all four series we observe substantial growth from 2016 through 2022, with particularly steep growth beginning in 2020. This growth trend is consistent across all groups and reflects Wagescape’s expanding coverage of the job market rather than an underlying increase in actual job openings. The parallel trends in the pre-SB-8 period and absence of a visible discontinuity at SB-8’s signing date support the identifying assumption.

Panel (c) shows posted salaries for RHP and non-RHP physicians in Texas and the control states. Table 1 presents summary statistics for the regression sample.

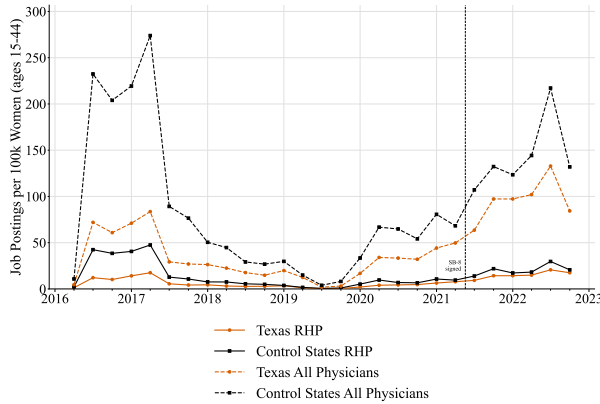
4.4 Balance: Texas vs. Control States vs. Other States

Table 2 presents pre-period (pre-May 2021) means for all NPI regression outcomes across three groups: Texas, the restricted control group (hostile/ban states), and all other states. The purpose of the balance check is to assess whether the restricted control group is a more credible counterfactual for Texas physicians than the full pool of states.

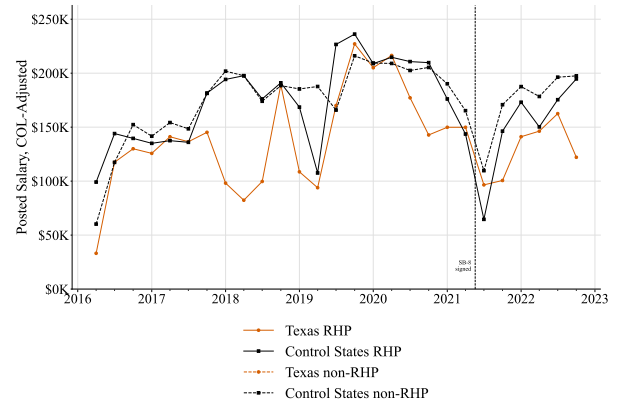
Texas is a large, growing state with a distinctive physician labor market. In the pre-period, Texas has lower counts of active RHPs per 100k women (120.3 per taxonomy group)



(a) Salary disclosure rate: share of postings with salary listed.



(b) Total job postings per 100k women: all physicians and RHP.



(c) Posted salary (COL-adjusted): RHP and non-RHP.

Figure 2: Wagescape descriptive figures, all comparing Texas (orange) to the restricted control group (black), with solid lines for RHP and dashed lines for all physicians. Panel (a): quarterly share of postings with salary disclosed; the four series track in parallel with no differential break around SB-8's signing, supporting the salary-sample validity assumption. Panel (b): total physician job postings per 100,000 women aged 15–44; post-2020 growth reflects Wagescape's expanding coverage rather than actual hiring. Panel (c): posted salary (BEA RPP cost-of-living adjusted) for RHP and non-RHP physicians in Texas and the restricted control states. The dashed vertical line in all panels marks SB-8's signing date (May 19, 2021). Source: Wagescape via Dewey Data Platform.

than both the restricted sample (157.0) and other states (161.5), reflecting Texas's rapid population growth relative to its physician supply. The restricted control group is closer to Texas on most flow outcomes (migration in and out, new entrants, retirements) than the full pool of other states, with differences roughly half as large for the restricted-sample comparison. This is the key motivation for using the restricted control group as the main specification: these states faced similar abortion-policy environments to Texas and exhibit more similar pre-period physician market trends, providing more credible parallel-trends

counterfactuals for the triple-difference identification. Full-sample (all-states) results are qualitatively unchanged.

	Texas	Restricted	Other	TX – Rest.	TX – Other
Active RHP	120.13	156.95	161.21	-36.82 (0.000)	-41.08 (0.000)
Active non-RHP	30.97	37.84	49.82	-6.87 (0.000)	-18.86 (0.000)
Salary, RHP (USD, COL-adj.)	155087	175164	163593	-20078 (0.000)	-8507 (0.024)
Salary, non-RHP (USD, COL-adj.)	163471	179526	169724	-16055 (0.000)	-6254 (0.010)
Postings/100k, RHP	5.91	17.74	21.51	-11.83 (0.000)	-15.60 (0.000)
Postings/100k, non-RHP	25.11	60.69	67.20	-35.58 (0.000)	-42.09 (0.000)

Table 2: **Balance: Texas compared to control group and other states.** Pre-period (before May 19, 2021) means for NPI regression outcomes across three groups: Texas, the restricted control group (hostile/ban states; 25 states), and all other states. All outcomes are per 100,000 women of childbearing age (ages 15–44). The “TX – Restricted” and “TX – Other” columns show simple mean differences; the restricted control group is closer to Texas on most flow outcomes, motivating its use as the main specification. Source: NPPES/CMS DAC.

5 Results

The triple-difference design exploits three sources of variation to identify the causal effect of SB-8’s civil liability enforcement mechanism. First, time: we compare the pre-signing period to the post-signing period (after May 19, 2021). Second, geography: we compare Texas to the restricted group of abortion-hostile control states. Third, specialty: we compare reproductive health physicians to all other physicians. The central identifying assumption is

the parallel trends assumption: absent SB-8, Texas RHPs would have evolved at the same rate as control-state RHPs, conditional on the common trajectory of non-RHP physicians within Texas. We also require no anticipation: physicians did not alter location decisions before SB-8’s signing.

$$\begin{aligned}
\text{Outcome}_{ist} = & \alpha + \beta_1 \cdot \underbrace{\text{Post SB8}_t}_{=1 \text{ if after May 19, 2021}} \times \underbrace{\text{Texas}_s}_{=1 \text{ if Texas}} \times \underbrace{\text{RHP}_i}_{=1 \text{ if Reprod. Health Phys.}} \\
& + \bar{\beta} \cdot \left[\text{Post SB8}_t \times \text{Texas}_s + \text{Post SB8}_t \times \text{RHP}_i + \text{Texas}_s \times \text{RHP}_i \right. \\
& \left. + \text{Post SB8}_t + \text{Texas}_s + \text{RHP}_i \right] + \underbrace{u_{ist}}_{\substack{\text{FP SE} \\ \text{(by state)}}} \tag{1}
\end{aligned}$$

where

$$\begin{aligned}
\text{Outcome}_{ist} \in & \{\text{Total Active, Exit State, Enter State, New Entrances, Retirements,} \\
& \text{Salary, and Job Posting Volume}\}
\end{aligned}$$

The coefficient β_1 is the triple-difference estimator of interest: the differential change in outcomes for Texas RHPs, relative to non-RHP physicians in Texas and relative to RHPs in the control states, following SB-8’s signing. Non-RHP physicians absorb general Texas physician-market trends – including population-driven changes in the per-100k denominator, COVID-related demand shocks, and state-level supply shifts unrelated to reproductive health – while control-state RHPs absorb national specialty-level trends. The triple-difference structure isolates the component of any Texas RHP change that is attributable to neither confounder.

Because Texas is the only treated state ($N_1 = 1$), conventional cluster-robust and wild cluster bootstrap standard errors are invalid for the scalar DDD coefficient ([Ferman and Pinto, 2019](#)). We use the heteroskedasticity-corrected closed-form standard error of [Ferman and Pinto \(2019\)](#), which remains valid with a single treated unit. All migration outcomes are normalized per 100,000 women of childbearing age. Event studies use cluster-robust

standard errors clustered by state, which remain valid when many time-varying coefficients are estimated simultaneously.

5.1 Results: Migration

We present migration results in three stages, each corresponding to one of the event study figures below. The comparison across figures motivates the DDD specification and illustrates what a naive DiD would miss. Table 3 presents the DDD scalar estimates.

	Active	Exit State	Enter State	New Entrances	Retirements
Texas $\times$ Pass-SB8	−43.214*	0.957	1.365	2.434	0.109
	(18.599)	(1.773)	(1.888)	(16.016)	(0.574)
Reproductive $\times$ Pass-SB8	−58.236***	1.987***	2.736***	14.434**	0.335*
	(5.158)	(0.492)	(0.524)	(4.442)	(0.159)
β_1	27.648	−0.156	−0.745	−0.052	−0.050
	(25.720)	(1.175)	(1.184)	(8.727)	(0.212)
Unit FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes
Control Group	Restricted	Restricted	Restricted	Restricted	Restricted
Observations	3,120	3,120	3,120	3,120	3,120
Pre-period mean (TX RHP)	353.16	1.23	1.65	8.19	0.23

Table 3: Triple-difference estimates of the effect of SB-8 on physician outcomes per 100,000 women of childbearing age (NPI Panel, 2007–2022). Columns are the five outcome variables; the specification includes unit (state $\times$ group) and date fixed effects. β_1 is the triple-difference coefficient (Texas $\times$ RHP $\times$ Post-SB8). Ferman–Pinto (2019) heteroskedasticity-corrected SEs in parentheses. Control group: states classified as hostile or total-ban under the Dench–Lindo typology (25 states). Fixed-effect robustness across six specifications is in Appendix A. Source: NPPES/CMS DAC.

(*) $p < 0.10$; (**) $p < 0.05$; (***) $p < 0.01$

DiD: Treated physicians. Figure 3 compares Texas RHPs to control-state RHPs in a DiD event study. Texas entered the period with fewer active RHPs per 100,000 women of childbearing age than the control-state average, and that gap was narrowing in the pre-

period: pre-period coefficients slope upward, indicating convergence.⁶ An upward-sloping pre-period is inconsistent with the parallel trends assumption for a naive DiD of Texas RHPs against control-state RHPs: the convergence trend would continue absent any policy change, rendering a naive DiD estimate biased.

DiD: Control physicians. Figure 4 presents the same DiD for non-RHP physicians – those with no professional exposure to SB-8’s “aid or abet” provisions. The pre-period pattern closely mirrors Figure 3: Texas non-RHPs also exhibit upward pre-period convergence at a similar rate. Because the pattern appears in both the specialty group exposed to the CLEM mechanism and the specialty group with no exposure, it cannot be attributed to SB-8. It reflects a general Texas physician-market trend. Attributing this trend to treatment would produce a biased estimate of the CLEM effect.

Triple-difference estimates. The DDD design addresses this directly by differencing out the general Texas physician-market trend identified in Figure 4. Figure 5 shows the resulting DDD event study. Pre-period coefficients are attenuated relative to either DiD figure; the parallel trends assumption is more credibly satisfied, though a modest residual pre-period slope in the active count panel remains.⁷ No outcome shows a post-signing break or drift.

Table 3 presents the scalar DDD estimates. For the primary outcome – total active RHP physicians – $\hat{\beta}_1 = 27.648$ per 100,000 women of childbearing age, equivalent to approximately 7.8 percent of the pre-period Texas RHP mean of 353.2.⁸ Interstate migration flows are similarly indistinguishable from zero: $\hat{\beta}_1 = -0.156$ for exit from Texas and $\hat{\beta}_1 = -0.745$ for entry. Career-boundary transitions are near zero as well: $\hat{\beta}_1 = -0.052$ for new career

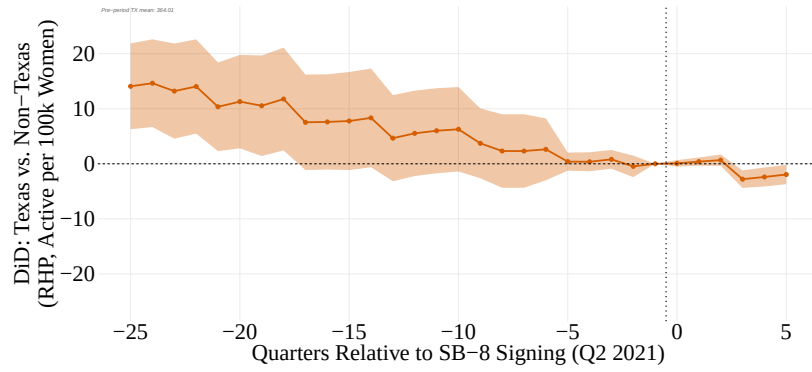
⁶This convergence is a denominator artifact: Texas physician counts grow in absolute terms, but the female population aged 15–44 grows faster, initially suppressing the per-100k stock. Running the same event study on raw physician counts reverses the pre-period slope, confirming that the trend reflects population dynamics rather than physician behavior.

⁷The residual pre-period slope in the active physician stock motivates the synthetic estimators in Section 5.2, which construct a counterfactual for Texas without imposing parallel pre-trends.

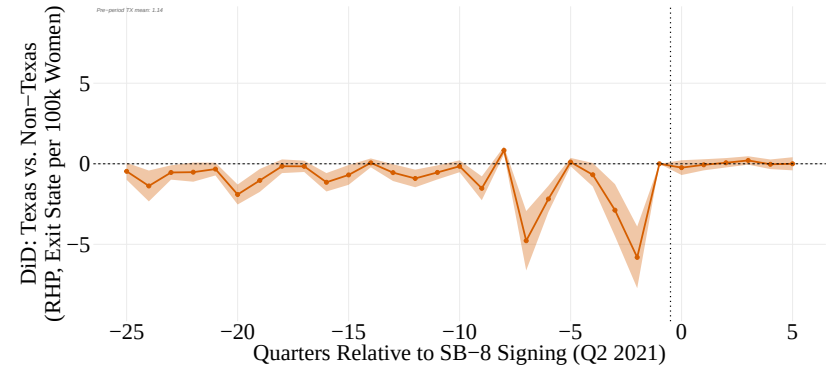
⁸The pre-period mean is computed from the collapsed regression panel, in which all RHP taxonomy cells within a state-quarter are summed to a single group observation. This differs from the per-taxonomy-cell means of 120.3 reported in Table 2.

entrances and $\hat{\beta}_1 = -0.050$ for retirements. Standard errors for all outcomes are reported in Table 3. We find no differential change in interstate migration, new career entrances, or retirements for reproductive health physicians in Texas following SB-8's signing. Section 5.2 examines whether posted wages and job posting volume show a comparable pattern.

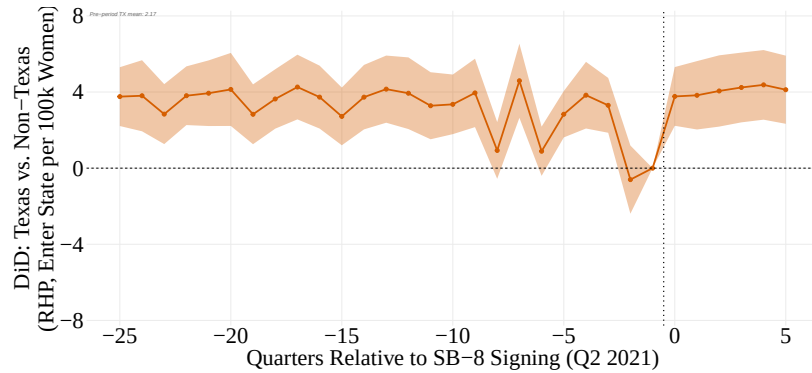
Diff-in-Diff Estimates: Treated Physicians



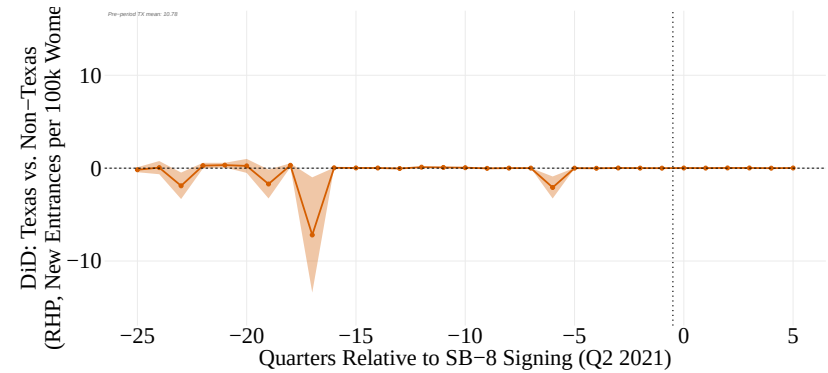
(a) Total active RHPs.



(b) Exit from Texas.



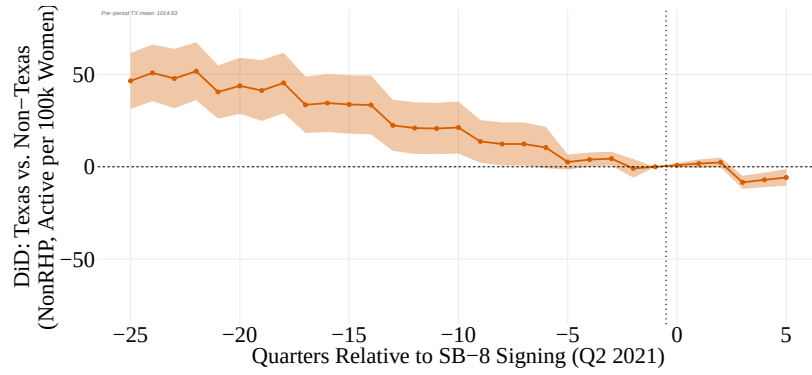
(c) Entry to Texas.



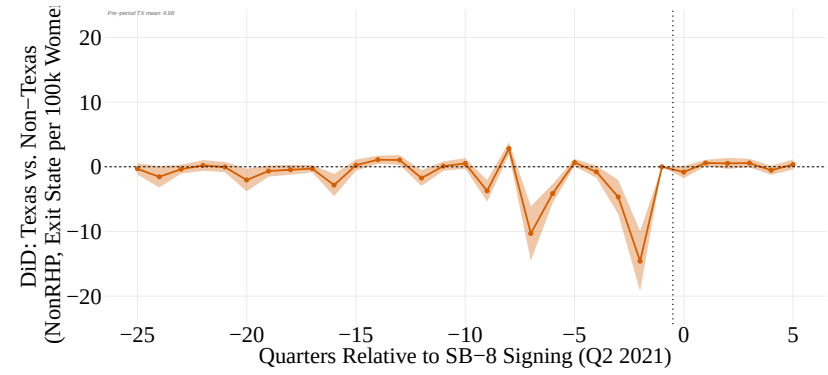
(d) New career entrances.

Figure 3: Difference-in-differences event study estimates for reproductive health providers (RHP: OB/GYN, Family Medicine, and Emergency Medicine), comparing Texas RHP physicians to non-Texas RHP physicians over time. The estimating equation includes state and date fixed effects; the reference period is Q1 2021 (event time -1). Each coefficient represents the quarterly Texas–non-Texas differential for RHP physicians relative to the reference quarter. The four panels show: total active RHPs per 100k women, out-migration (exit from Texas), in-migration (entry to Texas), and new career entrances. Sample: 2015–2022 quarterly, restricted control group; 95% confidence intervals shown. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

Diff-in-Diff Estimates: Control Physicians



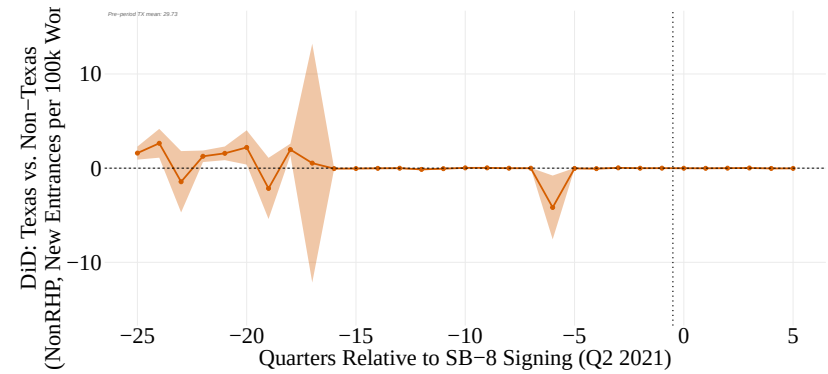
(a) Total active non-RHPs.



(b) Exit from Texas.



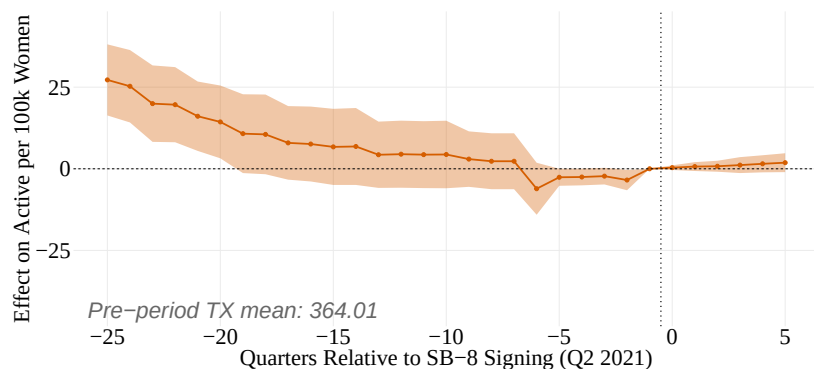
(c) Entry to Texas.



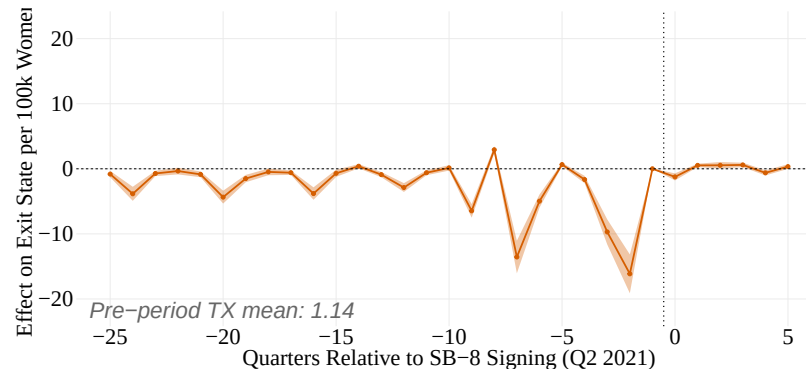
(d) New career entrances.

Figure 4: Difference-in-differences event study estimates for non-reproductive-health physicians (the control group), comparing Texas non-RHP physicians to non-Texas non-RHP physicians over time. The estimating equation includes state and date fixed effects; the reference period is Q1 2021 (event time -1). The four panels show: total active non-RHPs per 100k women, out-migration (exit from Texas), in-migration (entry to Texas), and new career entrances. These figures serve as a placebo check: absent SB-8 exposure, the Texas differential should be flat throughout. Sample: 2015–2022 quarterly, restricted control group; 95% confidence intervals shown. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

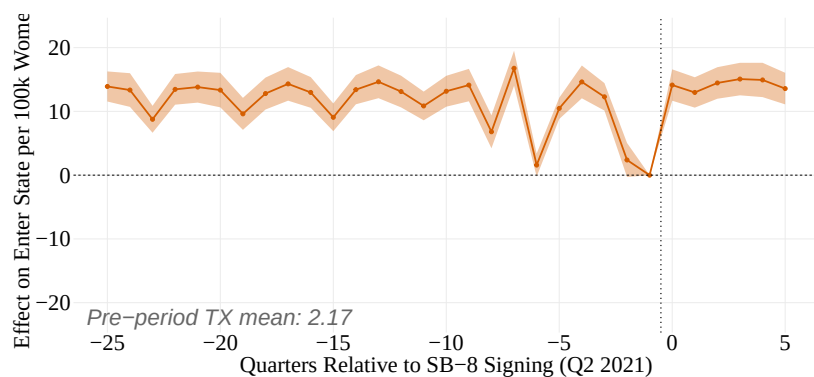
Triple-Difference Estimates: All Reproductive Health Providers



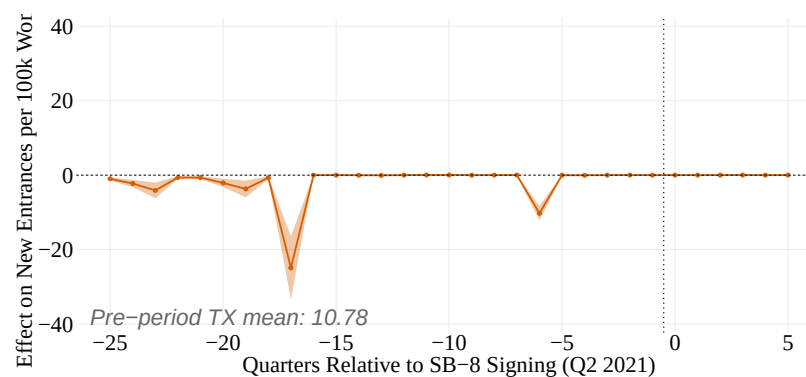
(a) Total active RHPs.



(b) Exit from Texas.



(c) Entry to Texas.



(d) New career entrances.

Figure 5: Triple-difference event study estimates for reproductive health provider (RHP: OB/GYN, Family Medicine, and Emergency Medicine) outcomes. The estimating equation is $\text{Outcome}_{ist} = \alpha + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[\text{event_time} = k] \times \text{Texas}_s \times \text{RHP}_i + [\text{FEs and lower-order terms}] + \varepsilon_{ist}$. Coefficients β_k are quarterly treatment effects relative to Q1 2021 (reference period: event time -1). The four panels show: total active RHPs, out-migration (exit from Texas), in-migration (entry to Texas), and new career entrances. All outcomes are per 100,000 women of childbearing age. Unit (state $\times$ group) and date fixed effects are included. Sample: 2015–2022 quarterly, restricted control group. Shaded regions show 95% confidence intervals. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

5.2 Results: Job Postings and Wage

The central identification challenge in the wage analysis is that posting-level salary data are sparse and heteroskedastic: cell sizes range from a handful of postings in small states and early quarters to thousands in large states and later years. With a single treated state, this heteroskedasticity invalidates conventional cluster-robust inference. We address this using the Ferman–Pinto (2019) heteroskedasticity-corrected closed-form standard error, which accounts explicitly for cell-size variation. The treated group is defined consistently with the CLEM mechanism: OB/GYN, Emergency Medicine, and OB-doing Family Medicine (Family Medicine physicians whose job title indicates obstetric, emergency, or ambiguous reproductive content). Panel A of Table 4 presents the full-sample results for three outcomes—log COL-adjusted posted salary, RHP job postings per 100,000 women of childbearing age, and COL-adjusted salary in USD levels—across six fixed-effect specifications. Columns progress from no fixed effects (1) to unit (state $\times$ group) and quarter-year fixed effects (6).

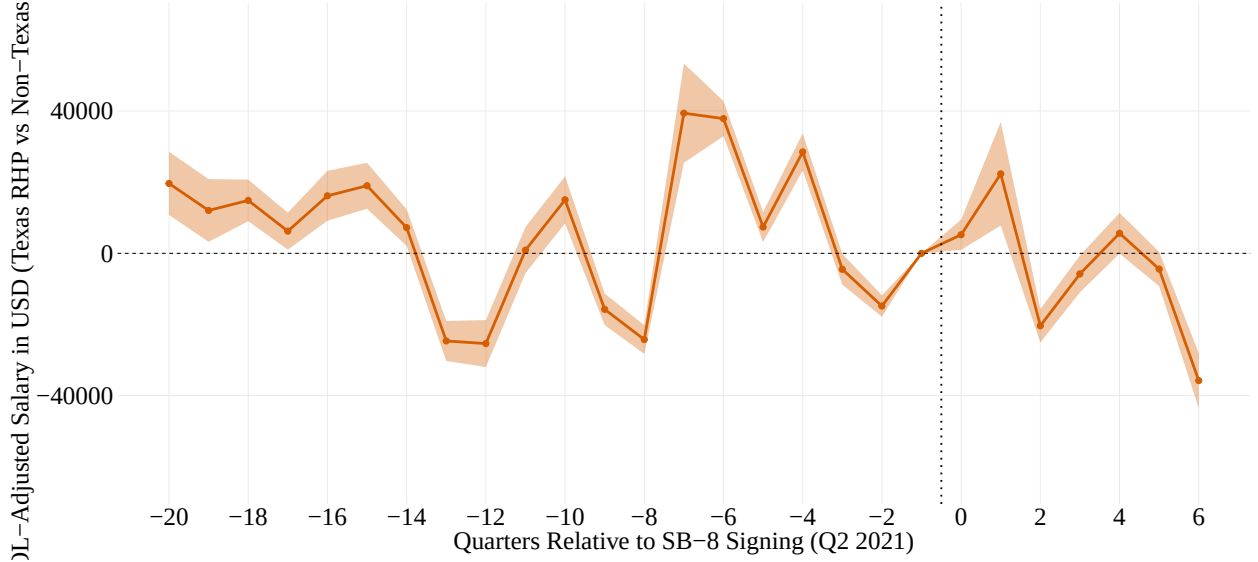
All three outcomes are statistically indistinguishable from zero in the preferred specification (Column 6). The COL-adjusted salary coefficient is \$332; the log salary coefficient is -0.034 , equivalent to approximately -3.4 percent; and the postings-per-100k coefficient is -2.702 . Standard errors and significance levels are reported in Table 4. The Ferman–Pinto (2019) heteroskedasticity-corrected standard errors are large because posting-level data generate severe cell-size heteroskedasticity, and conventional cluster-robust inference is invalid with a single treated state. The event study in Figure 6 shows no post-SB-8 break in either salary or posting volume, and pre-period coefficients are statistically indistinguishable from zero throughout the sample window, consistent with the parallel trends assumption.

Wage Triple-Difference Estimates by Employment Type and Fixed-Effect Specification						
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Total (Full Sample)</i>						
Log Salary	-0.187 (0.157)	-0.180 (0.161)	-0.145 (0.185)	-0.146 (0.185)	0.011 (0.133)	-0.034 (0.138)
Postings/100k	-35.448 (22.435)	-33.736 (21.186)	16.024 (38.114)	17.315 (39.302)	-11.864 (20.896)	-2.702 (16.002)
Salary (USD, COL-adj.)	-25730 (23004)	-23624 (22688)	-11885 (24633)	-11105 (24998)	4258 (22647)	332 (22667)
Pre-period mean (TX RHP): Log Salary			11.612			
Pre-period mean (TX RHP): Postings/100k			5.40			
Pre-period mean (TX RHP): COL Salary (USD)			141042			
<i>Panel B: Full-Time (Non-Locum)</i>						
Log Salary	-0.199 (0.174)	-0.192 (0.178)	-0.156 (0.199)	-0.158 (0.197)	-0.011 (0.152)	-0.044 (0.160)
Postings/100k	-33.964** (15.487)	-32.660** (14.444)	7.539 (30.239)	8.585 (31.239)	-16.488 (17.566)	-9.070 (11.812)
Salary (USD, COL-adj.)	-28981 (26196)	-26861 (25955)	-15286 (27697)	-14985 (27780)	-1286 (25894)	-2384 (26062)
Pre-period mean (TX RHP): Log Salary			11.599			
Pre-period mean (TX RHP): Postings/100k			4.84			
Pre-period mean (TX RHP): COL Salary (USD)			139654			
<i>Panel C: Locum Tenens</i>						
Log Salary	0.132 (0.154)	0.130 (0.156)	0.118 (0.150)	0.125 (0.160)	0.178 (0.206)	0.045 (0.168)
Postings/100k	-0.503 (9.978)	-0.255 (9.819)	8.950 (8.559)	9.431 (8.704)	5.062 (8.611)	7.120 (8.008)
Salary (USD, COL-adj.)	39633 (43170)	40104 (42729)	44243 (29112)	49438* (25228)	60117* (33377)	37285 (27623)
Pre-period mean (TX RHP): Log Salary			12.165			
Pre-period mean (TX RHP): Postings/100k			0.72			
Pre-period mean (TX RHP): COL Salary (USD)			199383			
State FE	No	Yes	Yes	Yes	—	—
Year FE	No	No	Yes	Yes	Yes	—
Quarter-of-Year FE	No	No	No	Yes	Yes	—
State $\times$ Group FE	No	No	No	No	Yes	Yes
Quarter $\times$ Year FE	No	No	No	No	No	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes	Yes

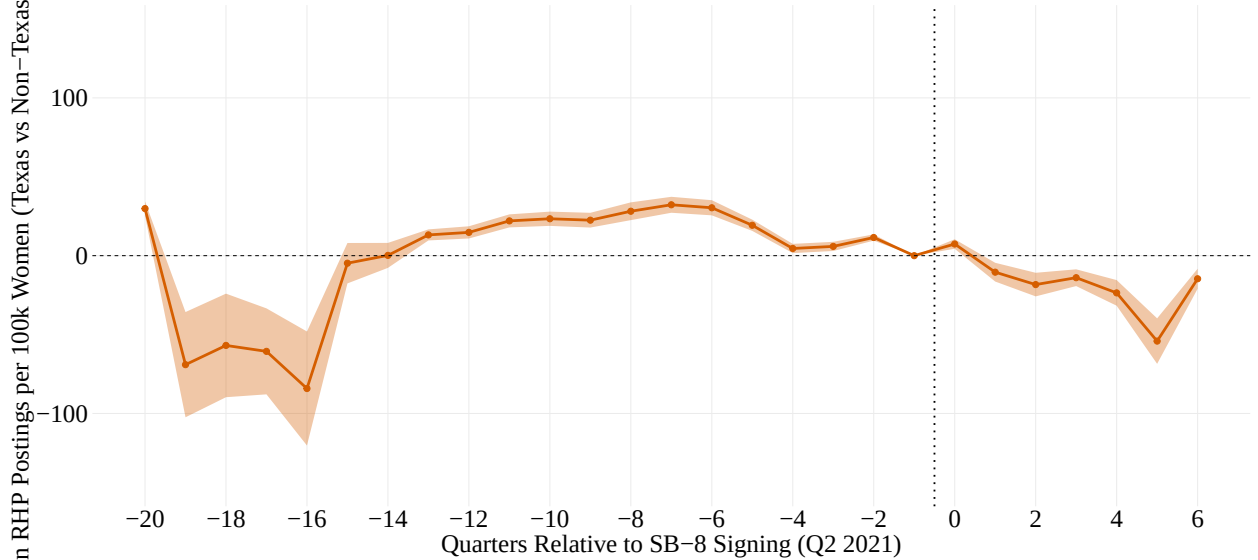
Table 4: Wage triple-difference β_1 estimates across employment-type subsamples and six fixed-effect specifications (columns). Panel A is the full sample; Panel B restricts to full-time (non-locum) postings; Panel C shows locum tenens postings only (identified by “locum” in job title, $\approx 10\%$ of postings). The Postings outcome is postings per 100,000 women of childbearing age. Columns (5)–(6) use state $\times$ group (RHP/non-RHP) unit fixed effects. Ferman–Pinto (2019) heteroskedasticity-corrected SEs (M = posting count). Control group: Dench–Lindo hostile/ban states (25 states).

Wage Triple-Difference Estimates by Geography and Fixed-Effect Specification						
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Total (Full Sample)</i>						
Log Salary	-0.187 (0.157)	-0.180 (0.161)	-0.145 (0.185)	-0.146 (0.185)	0.011 (0.133)	-0.034 (0.138)
Postings/100k	-35.448 (22.435)	-33.736 (21.186)	16.024 (38.114)	17.315 (39.302)	-11.864 (20.896)	-2.702 (16.002)
Salary (USD, COL-adj.)	-25730 (23004)	-23624 (22688)	-11885 (24633)	-11105 (24998)	4258 (22647)	332 (22667)
Pre-period mean (TX RHP): Log Salary			11.612			
Pre-period mean (TX RHP): Postings/100k			5.40			
Pre-period mean (TX RHP): COL Salary (USD)			141042			
<i>Panel B: Urban</i>						
Log Salary	-0.144 (0.289)	-0.137 (0.294)	-0.093 (0.328)	-0.093 (0.327)	-0.013 (0.198)	-0.038 (0.192)
Postings (Count)	-3144*** (705)	-3155*** (712)	-2710*** (534)	-2697*** (533)	-2035*** (598)	-1964*** (569)
Salary (USD, COL-adj.)	-16849 (31167)	-15072 (31999)	-3564 (38868)	-3203 (39029)	1836 (28605)	1673 (28416)
Pre-period mean (TX RHP): Log Salary			11.541			
Pre-period mean (TX RHP): Postings (Count)			213.19			
Pre-period mean (TX RHP): COL Salary (USD)			132593			
<i>Panel C: Rural</i>						
Log Salary	-0.314* (0.170)	-0.299* (0.171)	-0.307* (0.175)	-0.324* (0.174)	-0.275 (0.184)	-0.341** (0.170)
Postings (Count)	-139 (105)	-140 (106)	-73.430 (55.461)	-71.764 (55.288)	-43.244 (83.021)	-37.312 (75.410)
Salary (USD, COL-adj.)	-58974 (40785)	-55940 (40158)	-51546 (39626)	-53509 (39621)	-46241 (40608)	-51602 (39832)
Pre-period mean (TX RHP): Log Salary			11.921			
Pre-period mean (TX RHP): Postings (Count)			22.90			
Pre-period mean (TX RHP): COL Salary (USD)			177442			
State FE	No	Yes	Yes	Yes	—	—
Year FE	No	No	Yes	Yes	Yes	—
Quarter-of-Year FE	No	No	No	Yes	Yes	—
State × Group FE	No	No	No	No	Yes	Yes
Quarter × Year FE	No	No	No	No	No	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes	Yes

Table 5: Wage triple-difference β_1 estimates across geographic subsamples and six fixed-effect specifications (columns). Panel A is the full sample; Panels B and C split by posting zip code urbanicity (USDA RUCA 2020); zip codes with no RUCA match ($\approx 30\%$ of postings) are excluded from Panels B–C. The Postings outcome in Panel A is postings per 100,000 women of childbearing age; in Panels B–C it is raw posting count (levels), since the per-100k denominator is state-level and does not vary by urbanicity. Columns (5)–(6) use state $\times$ group (RHP/non-RHP) unit fixed effects. Ferman–Pinto (2019) heteroskedasticity-corrected SEs (M = posting count). Control group: Dench–Lindo hostile/ban states (25 states).



(a) COL-adjusted salary in USD: Texas RHP vs. non-Texas RHP triple-difference, quarterly.

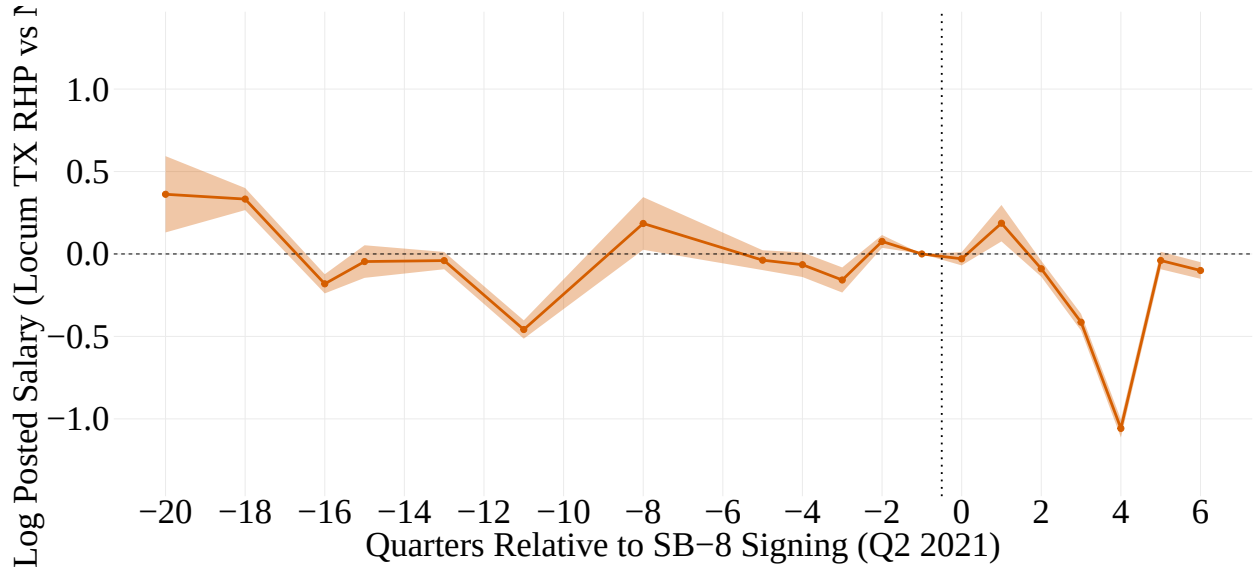


(b) RHP postings per 100k women: Texas vs. non-Texas triple-difference, quarterly.

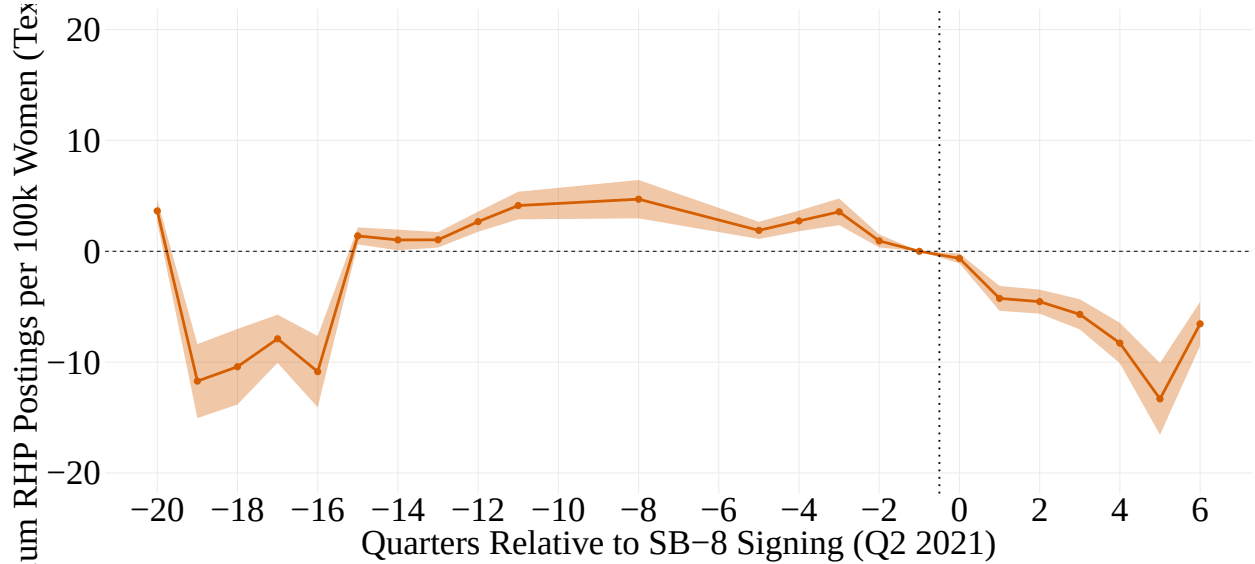
Figure 6: Triple-difference event study estimates for wage and posting outcomes using Wagescape data. Coefficients β_k are the quarterly treatment effects ($\text{TX} \times \text{RHP} \times \text{Post}$ interaction) relative to Q1 2021 (reference period: event time -1). Panel (a) uses BEA RPP cost-of-living adjusted posted salary in USD as the outcome; panel (b) uses treated RHP job postings per 100,000 women of childbearing age. Unit (state $\times$ group) and quarter-year fixed effects are included. Sample: Wagescape physician job postings, 2016–2022, restricted control group. Shaded regions show 95% confidence intervals. Pre-period coefficients are flat and the post-SB-8 period shows no break in either salary or posting volume. Cluster-robust SEs clustered by state. Source: Wagescape via Dewey Data Platform.

Locum Tenens Subsample. To examine a potentially more mobile segment of the physician labor market, we restrict the Wagescape sample to postings containing “locum” in the

job title ($N = 49,584$ salary observations; $N = 2,299$ state-quarter cells for the posting-volume outcome). Locum tenens physicians work on temporary assignments and face lower relocation costs than permanently placed physicians, making them a potentially more responsive margin to policy shocks like SB-8. Panel C of Table 4 presents triple-difference estimates for this subsample. The salary DDD estimator is statistically indistinguishable from zero, consistent with the null in the full sample. The posting-volume DDD is positive and statistically significant, indicating that Texas reproductive health providers saw a differential increase in locum postings per 100,000 women. This pattern is directionally consistent with a supply disruption: if permanent physicians reduced their Texas presence following SB-8, employers would need to fill gaps with temporary staff. The event study analogs are presented in Figure 7. We note that the posting-volume series is sensitive to thin state-quarter cells, and the positive DDD could also partly reflect differential growth in Wagescape’s locum coverage in Texas during this period; we interpret the posting result cautiously. Across both full-sample and locum subsamples, the wage evidence is consistent with the null migration result.



(a) Log posted salary: Texas RHP locum vs. non-Texas RHP locum differential, quarterly.



(b) Locum RHP postings per 100k women: Texas vs. non-Texas differential, quarterly.

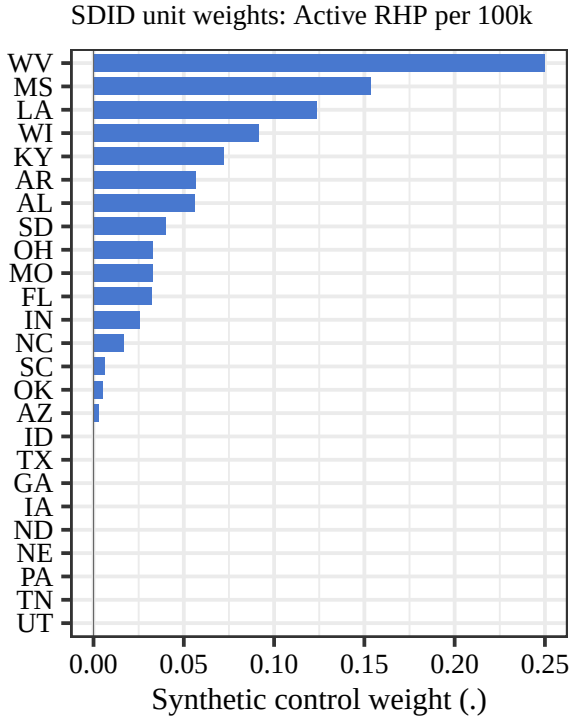
Figure 7: Triple-difference event study estimates for the locum tenens subsample. Coefficients β_k are the quarterly treatment effects ($\text{TX} \times \text{RHP} \times \text{Post}$) relative to Q1 2021 (reference period: event time -1). Panel (a) uses log BEA RPP cost-of-living adjusted posted salary; panel (b) uses locum RHP job postings per 100,000 women of childbearing age. State, year, and quarter fixed effects are included. Sample: Wagescape physician job postings with “locum” in the job title, 2016–2022, all U.S. states and D.C. Shaded regions show 95% confidence intervals. Cluster-robust SEs clustered by state. Source: Wagescape via Dewey Data Platform.

The wide confidence intervals in the wage specifications and the residual pre-period slope in the active physician stock both motivate synthetic estimators that relax the parallel trends assumption, which we turn to next.

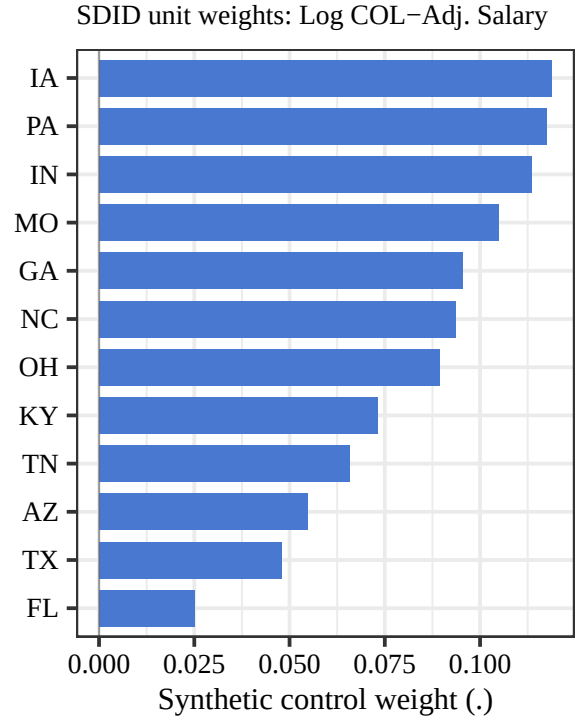
Synthetic Difference-in-Differences

The wide confidence intervals in the Wagescape DDD and the residual pre-period slope in the active physician stock event study motivate two synthetic estimators that do not impose parallel pre-trends. The Synthetic Difference-in-Differences (SDID) estimator of [Arkhangelsky et al. \(2021\)](#) constructs a reweighted donor pool of control states and a set of time weights jointly, so that the pre-period trajectory of synthetic Texas tracks both the level and trend of actual Texas. The Synthetic Control (SC) estimator weights donor states to match Texas’s pre-period outcome levels, without time-weighting. Both estimators operate on the RHP-only panel, using Texas as the single treated unit and the Dench-Lindo restricted states as the donor pool. Inference uses the placebo variance estimator with $B = 200$ replications. A tabular summary of all estimates is in [Appendix B](#).

Figure 8 shows the SDID unit weights (ω) for two representative outcomes. The weights are distributed across several control states rather than concentrated in one, indicating that Texas’s pre-period trajectory is well-approximated by a reweighted average of the donor pool and that the synthetic counterfactual does not depend heavily on any individual state.



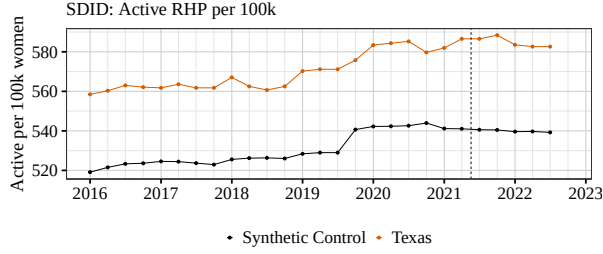
(a) NPI: Active physicians



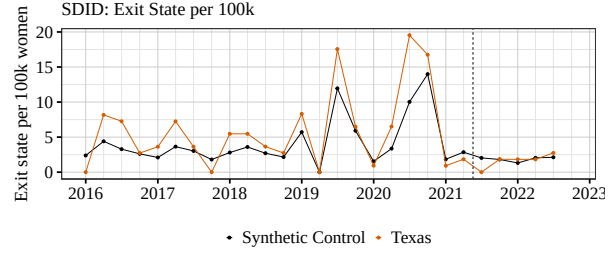
(b) Wagescape: Log salary

Figure 8: SDID unit weights (ω) for two representative outcomes. Each bar shows the weight assigned to a control state in constructing synthetic Texas. Weights are dispersed across the donor pool rather than concentrated in any single state. Control group: Dench-Lindo restricted states.

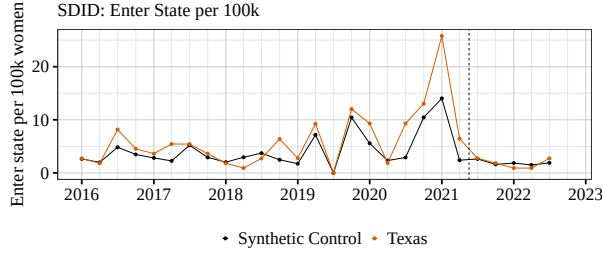
Figures 9 and 10 show the SDID trajectory plots: Texas (solid orange) against synthetic Texas (dashed black). SDID is designed to match the pre-period trend of Texas, and the synthetic counterfactual tracks the actual trajectory closely in each pre-period panel. Neither NPI nor Wagescape outcomes show a post-signing divergence between Texas and synthetic Texas.



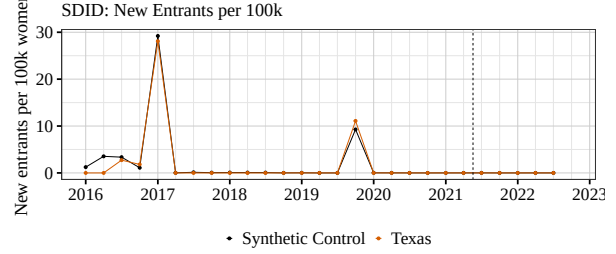
(a) Active RHP per 100k



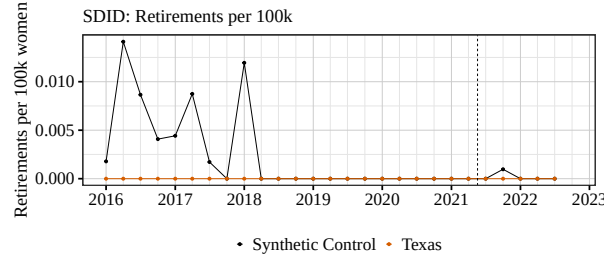
(b) Exit state per 100k



(c) Enter state per 100k

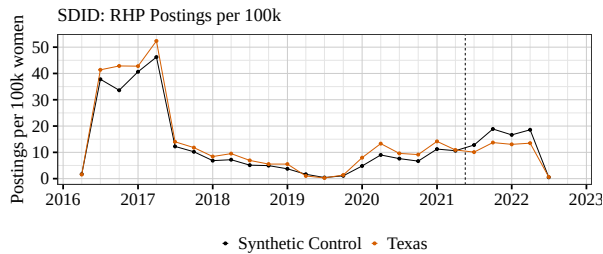


(d) New entrants per 100k

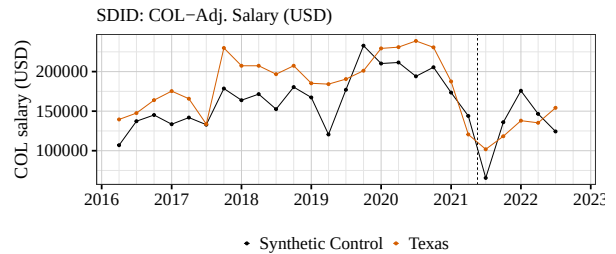


(e) Retirements per 100k

Figure 9: SDID trajectory plots for NPI outcomes. Texas (orange) versus synthetic Texas (black). SDID constructs the synthetic counterfactual to match the pre-period *trend* of Texas's RHP trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Shaded regions show 95% confidence intervals from placebo variance ($B = 200$). Control group: Dench-Lindo restricted states. Source: NPPES NPI registry.



(a) RHP postings per 100k



(b) COL-adjusted salary (USD)

Figure 10: SDID trajectory plots for Wagescape outcomes. Texas (orange) versus synthetic Texas (black). SDID constructs the synthetic counterfactual to match the pre-period *trend* of Texas's wage and posting trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Shaded regions show 95% confidence intervals from placebo variance ($B = 200$). Some Dench-Lindo states are excluded from the wage donor pool due to sparse salary coverage in early quarters. Source: Wagescape via Dewey Data Platform.

Figures 11 and 12 show the SC trajectory plots. Where SDID matches the pre-period trend of Texas, SC matches the pre-period levels: the synthetic Texas constructed by SC begins at the same outcome level as actual Texas without adjusting for time trends. The post-period null is equally apparent under SC. That the result is consistent whether the synthetic counterfactual is calibrated to levels or to trends indicates that the finding does not depend on the particular weighting scheme used to construct synthetic Texas.

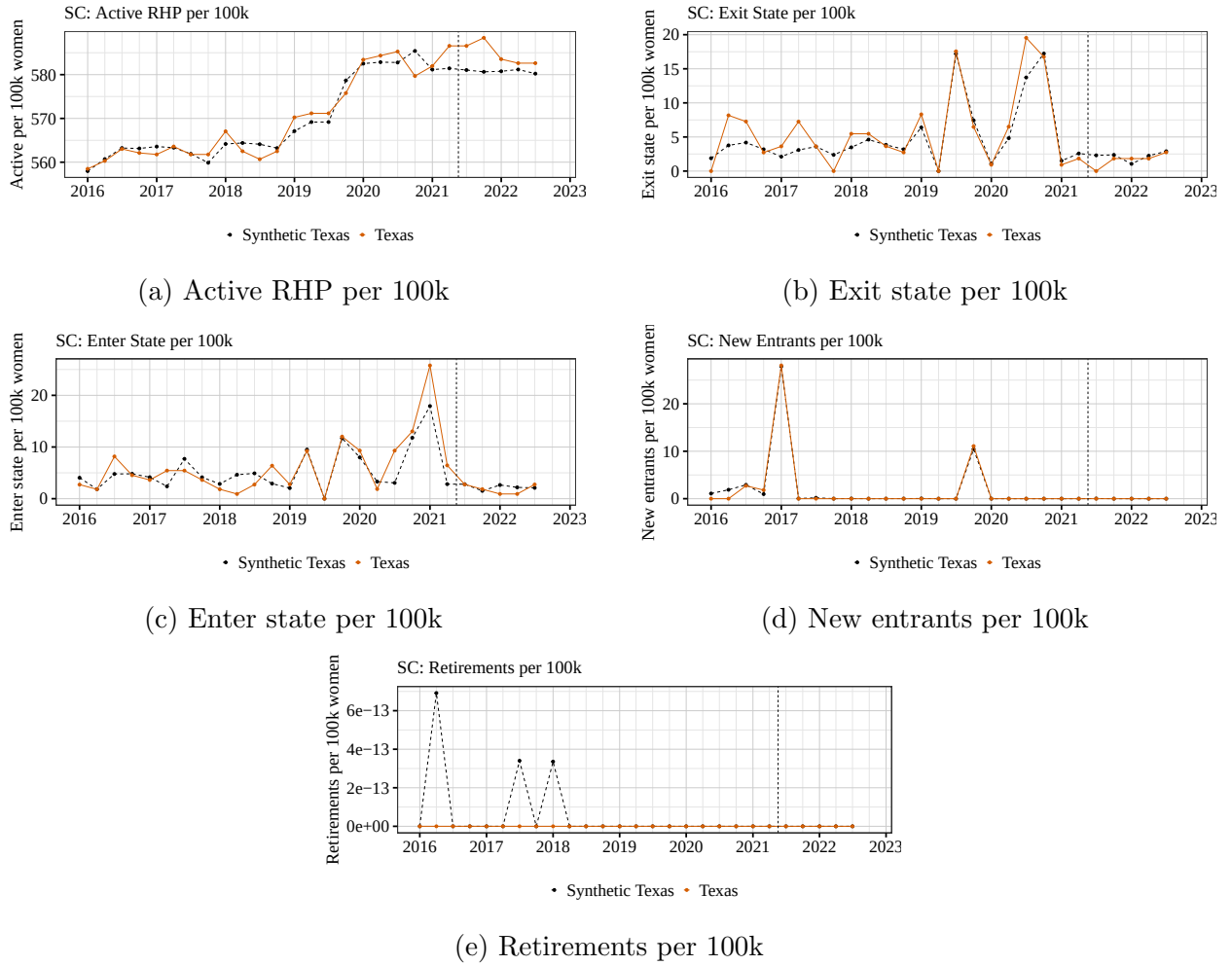
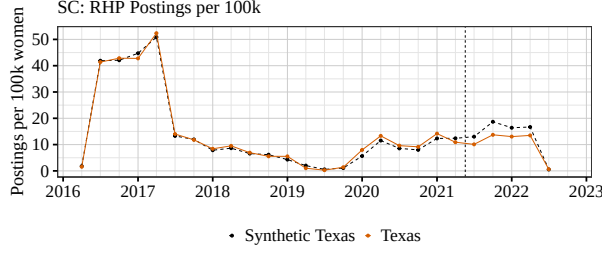
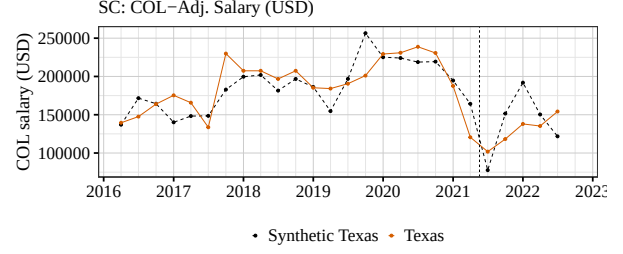


Figure 11: Synthetic control (SC) trajectory plots for NPI outcomes. Texas (solid orange) versus synthetic Texas (dashed black). SC constructs the synthetic counterfactual to match the pre-period *level* of Texas's RHP trajectory. That the null result is consistent across both SDID (trend-matched) and SC (level-matched) indicates it is not sensitive to the weighting scheme. Dashed vertical line marks SB-8 signing (May 19, 2021). Control group: Dench-Lindo restricted states. Source: NPPES NPI registry.



(a) RHP postings per 100k



(b) COL-adjusted salary (USD)

Figure 12: Synthetic control (SC) trajectory plots for Wagescape outcomes. Texas (solid orange) versus synthetic Texas (dashed black). SC constructs the synthetic counterfactual to match the pre-period *level* of Texas’s wage and posting trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Control group: Dench-Lindo restricted states. Source: Wagescape via Dewey Data Platform.

Figure 13 collects all estimates into a single coefficient plot. Columns are active physician stock (NPI) and the three Wagescape outcomes; rows are identification strategies. For Wagescape outcomes, the upper section additionally shows the DDD estimate under five alternative fixed-effect specifications (corresponding to columns 1–5 of Tables 4 and 5), from no fixed effects to the preferred unit-and-quarter-by-year specification; all DDD rows use Ferman–Pinto (2019) standard errors. The active stock column shows only the main DDD and synthetic estimators. Point estimates and 95% confidence intervals are centered on zero and statistically indistinguishable from zero across all outcomes and all estimators. We find no detectable change in migration, salary, or posting volume for reproductive health physicians in Texas following SB-8. Appendix Figure 16 extends the display to all five NPI migration outcomes.

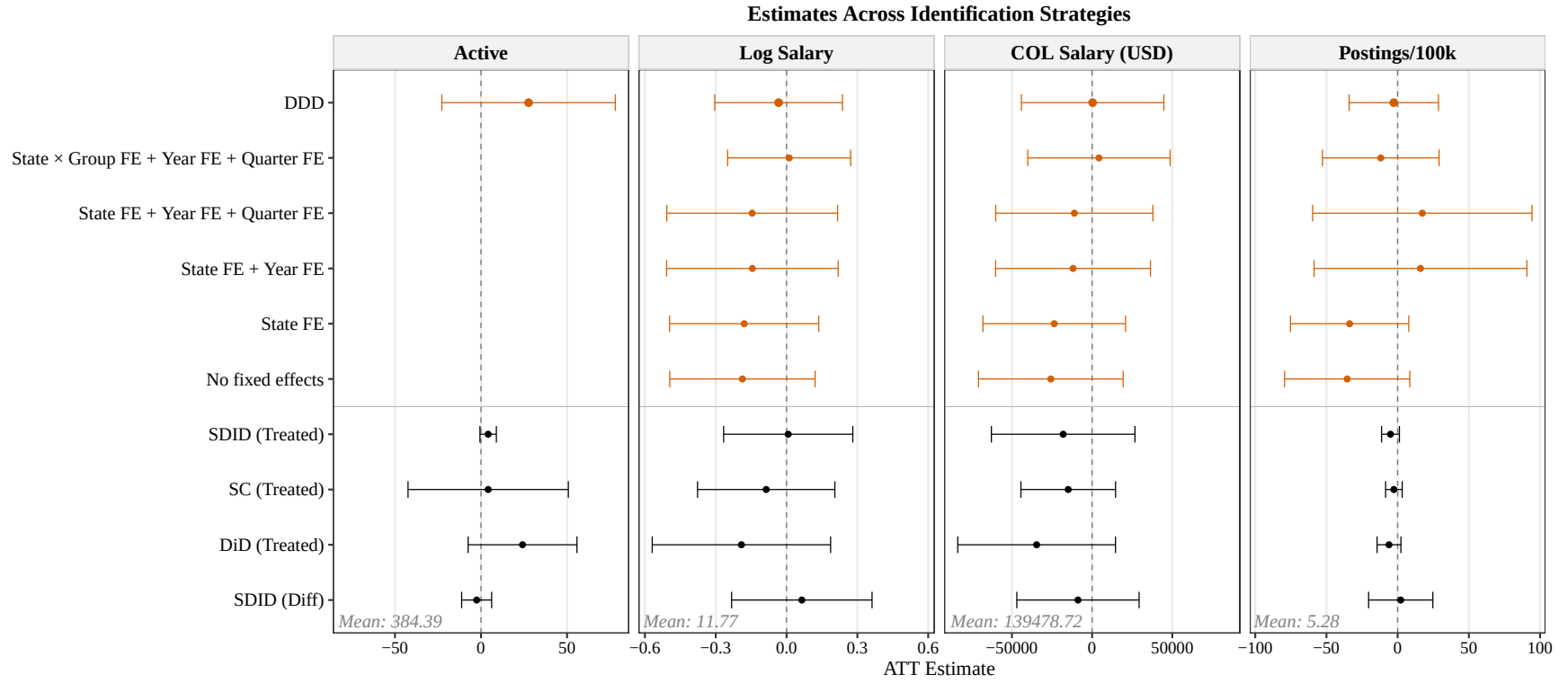


Figure 13: Coefficient plot. Columns: active physician stock (NPI, per 100k women), log posted salary, COL-adjusted salary (USD), and RHP postings per 100k (Wagescape). Rows: DDD with Ferman–Pinto (2019) SE (top), DDD under five alternative fixed-effect specifications for Wagescape outcomes only (all FP SE; wage FE rows are blank for the active stock column), SDID on treated physicians, SC on treated physicians, DiD on treated physicians, and SDID on the RHP – Non-RHP difference. SDID/SC/DiD use placebo variance ($B = 200$). X-axes are centered on zero; gray italic text shows the pre-period Texas RHP mean. Control group: Dench-Lindo restricted states.

6 Discussion

This paper examines whether SB-8’s civil liability enforcement mechanism altered the location decisions and compensation of reproductive health physicians, exploiting variation across physician specialties, states, and time in a triple-difference design. We find no statistically significant effect on any outcome we examine.

In the NPI migration data, the triple-difference estimate is statistically indistinguishable from zero for all five outcomes—total active physicians, interstate exit, interstate entry, new career entrants, and retirements. We find no differential change in exit rates or career entrances consistent with a deteriorating practice environment following SB-8’s signing on May 19, 2021.

In the Wagescape data, the evidence is equally consistent. The log salary triple-difference is close to zero (-0.034): reproductive health providers in Texas did not command a compensating wage differential for CLEM exposure, nor did they experience a differential wage decline. The same null appears in the dollar-value COL-adjusted salary specification and in the posting-volume outcome. The locum tenens subsample shows a positive and statistically significant effect on posting volume, which could indicate temporary staffing substitution; however, this result is also consistent with differential expansion of Wagescape’s locum coverage in Texas during this period, and we interpret it cautiously.

The null results are consistent across eight outcomes, two identification strategies, and a restricted control group of similarly abortion-hostile states. One interpretation is that the marginal expected cost of CLEM exposure was insufficient to overcome the substantial financial and professional costs of interstate relocation for established physicians, estimated at \$150,000–\$250,000 in licensing, credentialing, and transition costs. We cannot rule out alternative explanations, including that the pool of physicians most willing to exit in response to increased operating costs had already left Texas under prior TRAP regulations, or that the threat of CLEM litigation was too ambiguous to alter near-term location decisions.

Limitations and Next Steps

1. The retirement margin is poorly measured. Physicians do not deactivate their National Provider Identifiers when they retire, so the NPI panel substantially undercounts workforce exit. The retirement margin is particularly important for this analysis because physicians planning to retire in the near term face lower fixed costs of early exit than of relocation. As additional administrative data sources become available, including Medicare prescribing records and vital statistics restricted-access files linking birth certificates to provider identifiers, it will be possible to construct more precise measures of retirement and early career exit.
2. Given Texas’s restrictive abortion environment predating SB-8 (TRAP regulations), the physicians most willing to exit in response to a marginal increase in operating costs may have already left. The triple-difference design controls for pre-existing differential trends by comparing the change in RHP outcomes to the change in non-RHP outcomes within Texas, but cannot fully rule out that the pool of marginal movers was depleted before May 2021.
3. Heterogeneity by physician age is a natural extension. New physicians choosing an initial practice location bear no relocation cost and may be the most responsive margin; near-retirees face lower costs of early exit than of interstate relocation. As data on physician graduation year from CMS Physician Compare are linked to the NPI panel, it will be possible to examine whether younger and older cohorts exhibit differential responses.
4. County-level analysis would permit tests of whether physicians in blue Texas counties or low-relocation-cost border counties (e.g., Texarkana) are more responsive than those in rural or conservative counties, consistent with the prediction that physicians facing lower relocation costs should be more mobile.

5. A comparison of RHP physicians against specialty groups with no plausible exposure to SB-8 (such as ophthalmologists or rheumatologists) would provide an alternative falsification test for the DDD identification assumption.

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A NPI Triple-Difference: Fixed-Effect Robustness

Table 6 reports the triple-difference β_1 coefficient for all five NPI outcomes across six fixed-effect specifications, from no fixed effects (column 1) to saturated unit and date fixed effects (column 6). Because the analysis panel is balanced (26 states $\times$ 26 quarters $\times$ 2 groups, all cells observed), the Frisch–Waugh–Lovell theorem implies that any fixed effects orthogonal to the Texas $\times$ RHP $\times$ Post variation leave β_1 unchanged. All six columns therefore yield identical point estimates; IID standard errors are shown for cross-column comparability. Ferman–Pinto SEs are reported in Table 3.

	(1)	(2)	(3)	(4)	(5)	(6)
Active	27.648 (106.309)	27.648 (79.450)	27.648 (63.515)	27.648 (63.479)	27.648 (26.301)	27.648 (26.303)
Exit State	-0.156 (4.438)	-0.156 (4.333)	-0.156 (3.917)	-0.156 (3.841)	-0.156 (3.792)	-0.156 (2.508)
Enter State	-0.745 (4.639)	-0.745 (4.581)	-0.745 (3.997)	-0.745 (3.982)	-0.745 (3.965)	-0.745 (2.670)
New Entrances	-0.052 (53.196)	-0.052 (53.377)	-0.052 (23.170)	-0.052 (23.092)	-0.052 (23.146)	-0.052 (22.650)
Retirements	-0.050 (2.091)	-0.050 (2.099)	-0.050 (1.880)	-0.050 (1.825)	-0.050 (1.832)	-0.050 (0.812)
Observations	3,120	3,120	3,120	3,120	3,120	3,120
State FE	No	Yes	Yes	Yes	—	—
Year FE	No	No	Yes	Yes	Yes	—
Quarter-of-Year FE	No	No	No	Yes	Yes	—
State $\times$ Group FE	No	No	No	No	Yes	Yes
Quarter $\times$ Year FE	No	No	No	No	No	Yes
IID SE	Yes	Yes	Yes	Yes	Yes	Yes
Pre-period mean (TX RHP): Active	353.16	353.16	353.16	353.16	353.16	353.16
Pre-period mean (TX RHP): Exit State	1.23	1.23	1.23	1.23	1.23	1.23
Pre-period mean (TX RHP): Enter State	1.65	1.65	1.65	1.65	1.65	1.65
Pre-period mean (TX RHP): New Entrances	8.19	8.19	8.19	8.19	8.19	8.19
Pre-period mean (TX RHP): Retirements	0.23	0.23	0.23	0.23	0.23	0.23

Table 6: NPI triple-difference β_1 (Texas $\times$ RHP $\times$ Post-SB8) across five outcomes and six fixed-effect specifications. Because the collapsed panel is balanced, all six specifications yield identical point estimates (Frisch–Waugh–Lovell). “—” denotes a fixed effect subsumed by a higher-level FE. IID SEs shown; Ferman–Pinto SEs in Table 3. Control group: Dench–Lindo restricted states (25 states). Source: NPPES/CMS DAC.

(*) $p < 0.10$; (**) $p < 0.05$; (***) $p < 0.01$

B Synthetic Estimator Results

Table 7 reports point estimates and standard errors for the SDID, SC, and DiD estimators alongside the preferred DDD-FP estimates. Inference for the synthetic estimators uses the placebo variance estimator with $B = 200$ replications.

Table 7: SDID, SC, and DiD estimates for all eight outcomes alongside the preferred DDD-FP estimates from Tables 3 and 4. SDID, SC, and DiD operate on the RHP-only panel with Texas as the single treated unit; inference via placebo variance ($B = 200$). DDD-FP uses Ferman–Pinto (2019) heteroskedasticity-corrected SEs on the full RHP + non-RHP panel. Control group: Dench-Lindo restricted states. SEs in parentheses. NPI outcomes are per 100,000 women of childbearing age.

Outcome	SDID	SC	DiD-FP
<i>Panel A: NPI (per 100k women of childbearing age)</i>			
Active	4.145 (2.431)	4.170 (23.759)	23.774 (42.630)
Exit State	-1.472 (0.352)	-0.565 (0.266)	-0.156 (1.182)
Enter State	-1.338 (0.450)	0.259 (0.382)	-0.745 (1.181)
New Entrances	0.007 (0.013)	-0.006 (0.016)	-0.052 (8.904)
Retirements	0.000 (0.003)	0.000 (0.001)	-0.050 (0.215)
<i>Panel B: Wagescape Wage Outcomes</i>			
Log Salary	0.007 (0.140)	-0.086 (0.148)	-0.034 (0.138)
Postings per 100k	-5.022 (3.197)	-2.638 (2.972)	-2.702 (16.000)
COL Salary (USD)	-18,041 (22,837)	-14,886 (15,058)	332 (22,667)
Placebo variance for SDID and SC ($B = 200$; increase to $B = 1,000$ before publication). DiD-FP: Ferman–Pinto (2019) closed-form SE from Tables 3 and 4. SEs in parentheses.			

C SDID on the RHP – NonRHP Difference

As an additional robustness check, we apply SDID to the within-state, within-quarter difference between RHP and non-RHP outcomes. For each state s and quarter t , we compute $\Delta_{st} = y_{st}^{\text{RHP}} - y_{st}^{\text{NonRHP}}$ and then apply SDID with Texas as the single treated unit and the Dench-Lindo control states as the donor pool. This estimator replicates the triple-difference logic in the SDID framework: rather than netting out the non-RHP trend via an explicit interaction term, it does so by pre-differencing the outcome before constructing the synthetic counterfactual. Under the identifying assumption that the within-state RHP-minus-NonRHP gap would have evolved in parallel across states absent SB-8, this estimator recovers the same causal parameter as the DDD β_1 . Inference uses placebo variance with $B = 200$ replications. Figures 14 and 15 show trajectory plots; all estimates are null, consistent with the main results.

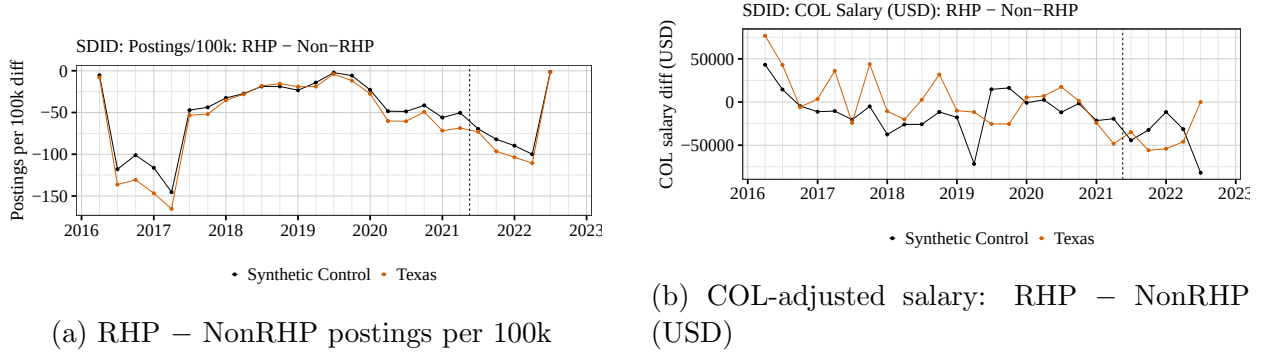
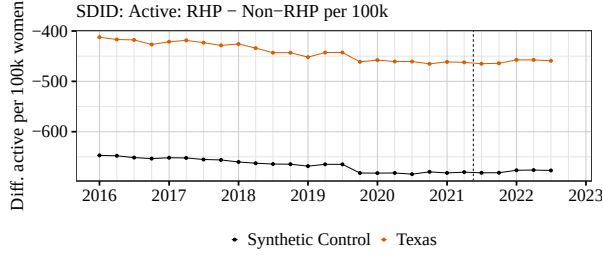
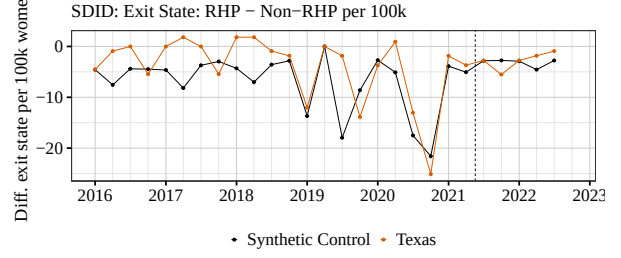


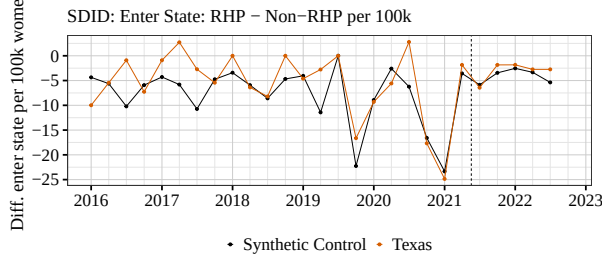
Figure 15: SDID trajectory plots on the (RHP – NonRHP) difference for Wagescape outcomes. Texas (solid orange) versus synthetic Texas (dashed black). The outcome is the within-state RHP-minus-NonRHP difference in posting volume or COL-adjusted salary, pre-differenced before applying SDID. Dashed vertical line marks SB-8 signing (May 19, 2021). Shaded regions show 95% confidence intervals from placebo variance ($B = 200$). Control group: Dench-Lindo restricted states. Source: Wagescape via Dewey Data Platform.



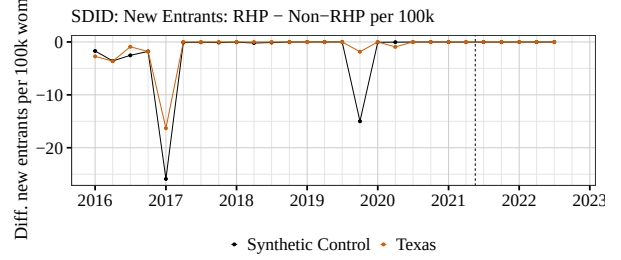
(a) Active: RHP – NonRHP per 100k



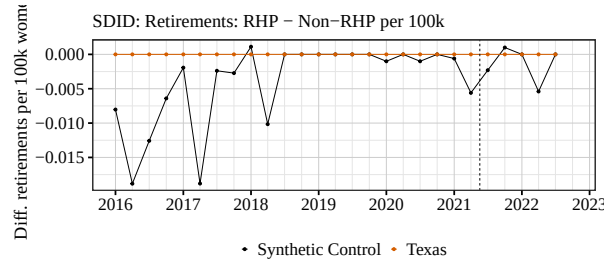
(b) Exit state: RHP – NonRHP per 100k



(c) Enter state: RHP – NonRHP per 100k



(d) New entrants: RHP – NonRHP per 100k



(e) Retirements: RHP – NonRHP per 100k

Figure 14: SDID trajectory plots on the (RHP – NonRHP) difference for NPI outcomes. Texas (solid orange) versus synthetic Texas (dashed black). The outcome is the within-state difference between RHP and non-RHP physician counts per 100,000 women of childbearing age, pre-differenced before applying SDID. Dashed vertical line marks SB-8 signing (May 19, 2021). Shaded regions show 95% confidence intervals from placebo variance ($B = 200$). Control group: Dench-Lindo restricted states. Source: NPPES NPI registry.

D Full NPI Migration Coefficient Plot

Figure 16 replicates the structure of Figure 13 for all five NPI migration outcomes. Each column is one outcome; rows are the same set of identification strategies. All estimates are null.

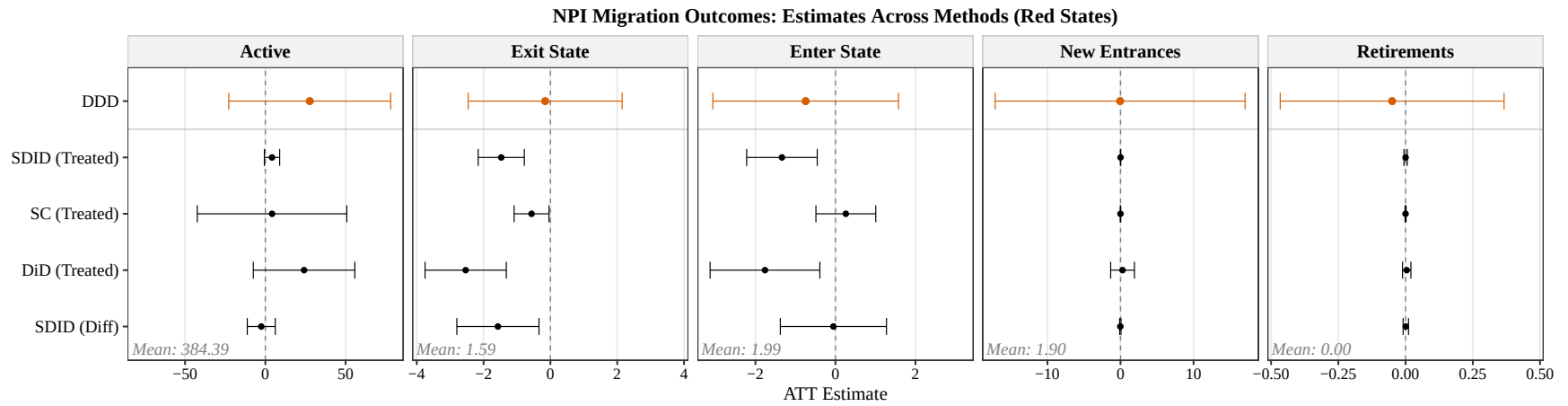


Figure 16: Coefficient plot for all five NPI migration outcomes (per 100,000 women of childbearing age). Columns: active stock, exit state, enter state, new career entrants, retirements. The four columns to the right should be interpreted with caution. Measurement error currently prevents precise estimation or interpretation of these coefficients. Rows: DDD (Ferman–Pinto SE), SDID on treated physicians, SC on treated physicians, DiD on treated physicians, and SDID on the (RHP – Non-RHP) difference. SDID/SC/DiD use placebo variance ($B = 200$). X-axes are centered on zero; gray italic text shows the pre-period Texas RHP mean. Control group: Dench-Lindo restricted states.